TEMCO LEARN

TStat8

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TStat8

The TStat8 thermostat: its specifications, the TStat8 software screen by screen, its hardware, and how to install and update it.

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TStat8 Manual

TStat8 Manual



User guide

June 2017

Introduction

This full-featured thermostat is designed for cooling and heating systems in residential and commercial buildings. The thermostat can be configured for use with air handlers,fan coils, VAV□s, modulating valves and many other HVAC applications. All models support BACnet and Modbus protocols which allows for easy integration with big name control systems like Niagara, Siemens,Honeywell, Johnson Controls, Delta, Reliable and Kreuter just to name a few.There are fie relays outputs and two analog outputs as well as 8 universal inputs. These can be confiured using the T3000 free software avialable for download. There are more than 300 settings with many options. This makes it possible to confiure these devices for most any HVAC application. Once the unit is confiured, you can save the fie and copy it to other controllers and backup project settings.Options are available for occupancy, zigbee wireless and humidity / enthalpy control.



The contents of Introduction

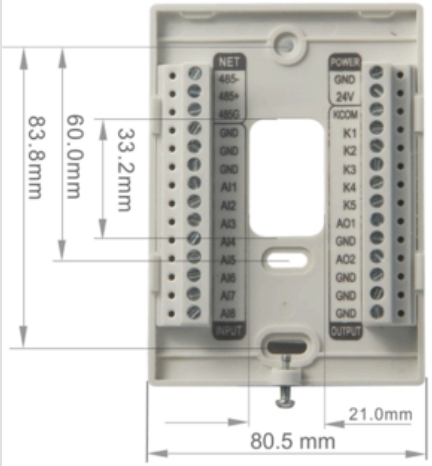
Specifications

Outputs	5 relay outputs 24vac @ 2 amps; 2 analog outputs 10V @100mA
8 Universal Inputs	10k therm, contacts, 4-20ma, 0-5V,0-10V
Operating range	-30~70°C(-22~158°F) / 0 to 99% RH
Supply voltage	12~24VAC/DC ±20%, 50-60Hz
Power consumption	100mA at 12VDC
Relay contacts	5 relays, 2A @ 24VAC UL File No.: E169380
Plastic Housing	Flammability rating UL 94 file E56070
Enclosure rating	IP31
Protocols	BACnet MS/TP and Modbus RTU RS485
Baudrate	9600, 19200, 38400, 57600, 76800, 115200
Temperature sensor	10K thermistor ±0.5°C
Setup Software	Free, no licensing, open source, down-load from website

Dimension



Rear



Backplate



Rear,backplate off



Front



Bottom

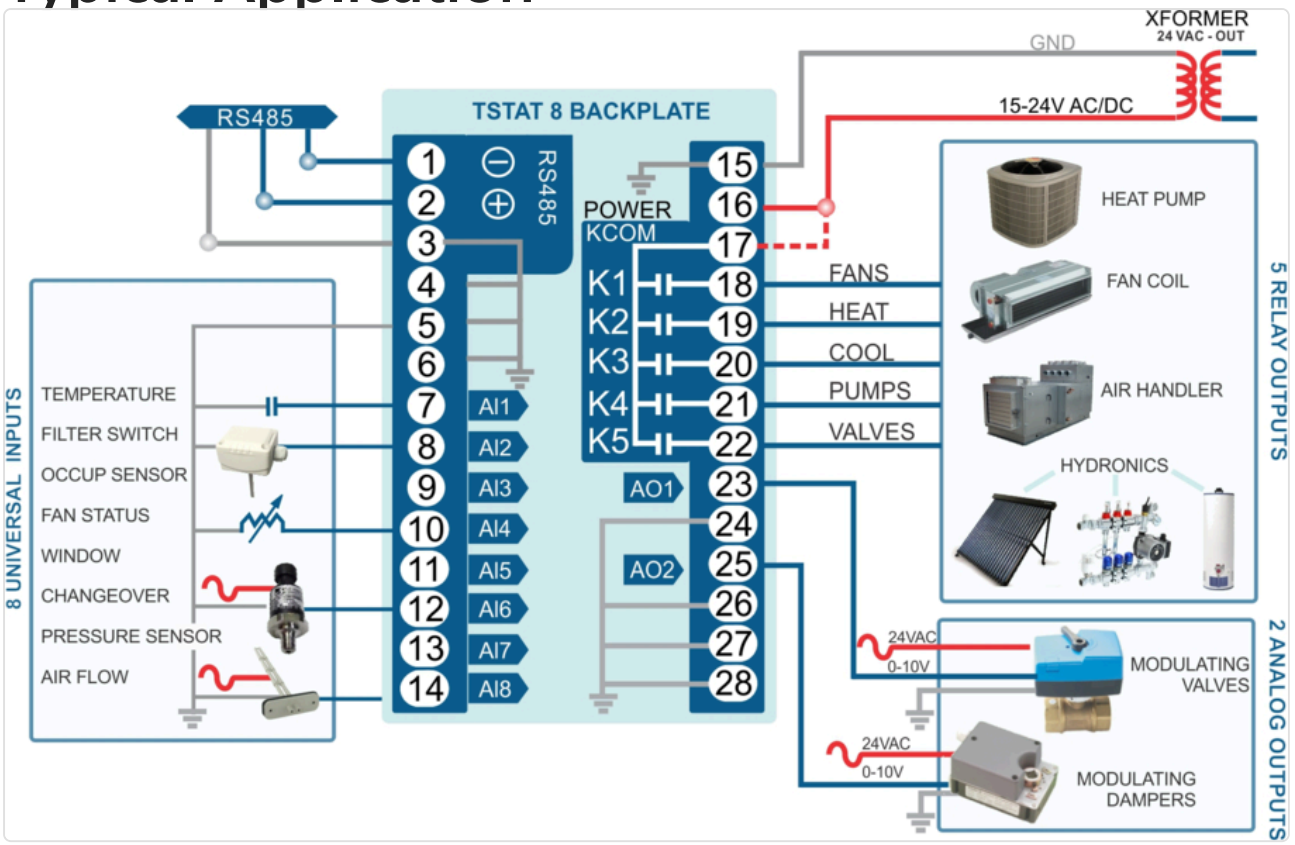
Part Number Scheme

Tstat8 – H

Code	Description
Tstat8	Thermostat

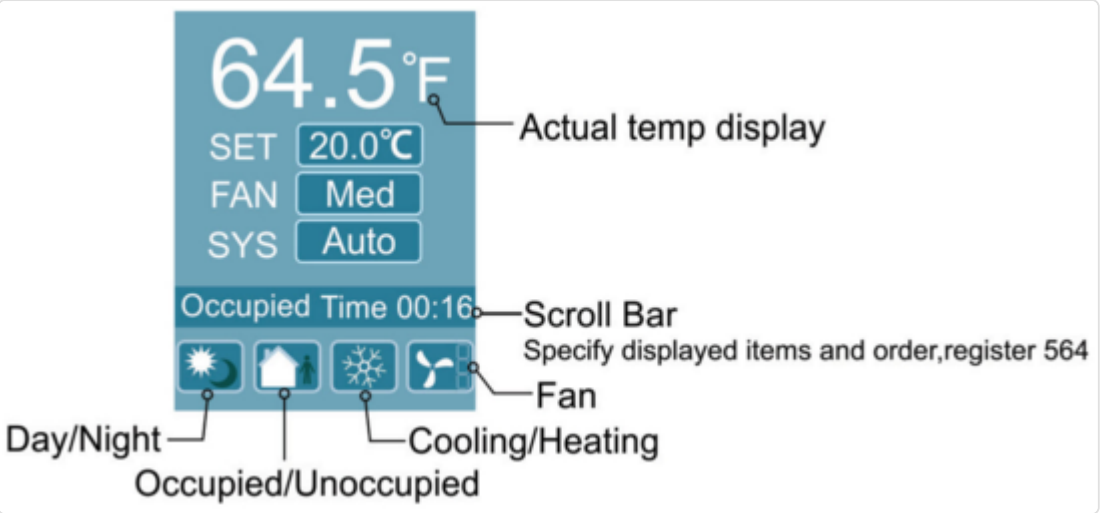
Code	Description
	Basic model temperature and clock
H	Humidity
OCC	Occupancy sensor
ZIGB	Zigbee

Typical Application



Menu Item Details

Tstat8 have several advanced menu items which can be adjusted in the fild to suit the specifi application and tune the operation of the thermostat. All the parameters are set up at the factory on an order-by-order basis and will give satisfactory results out of the box.



Tstat8 Software

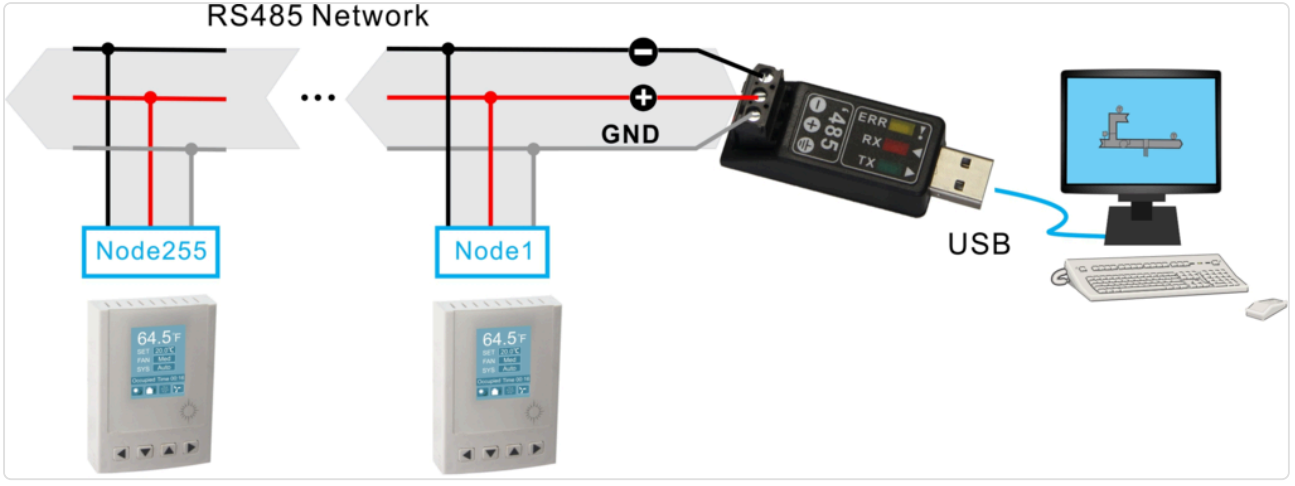
Hightlights:

-

- Baudrates of 1.2k,4.8k,9.6k,14.4k,19.2k, 38.4k, 57.6k, 76.8k and 115.2k.

- Well documented register list for easy integration with other systems.

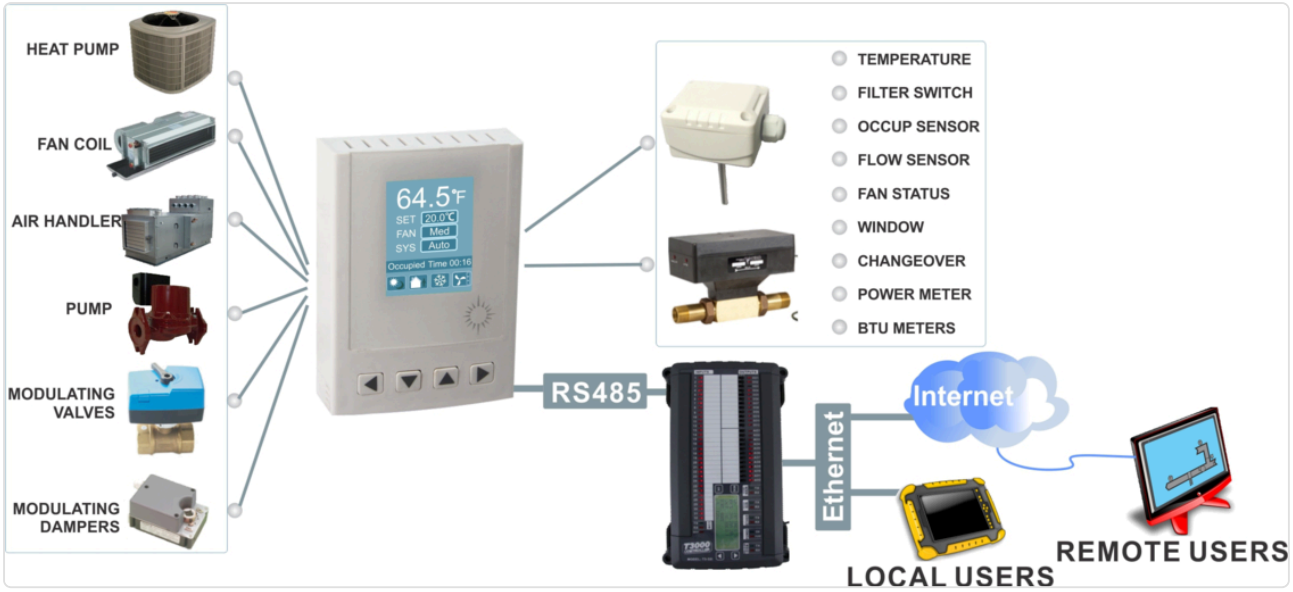
- Easily configure the thermostat for practically any application.



The contents of Tstat8 Software

T3000 Software

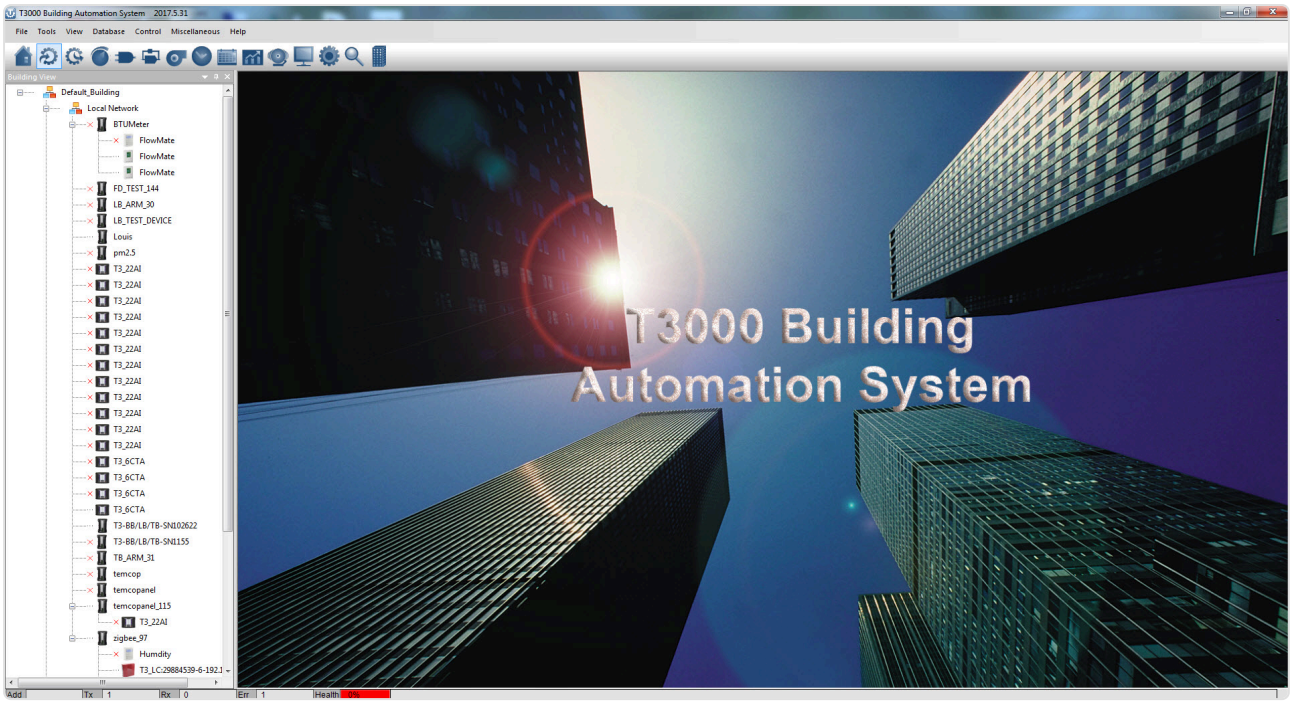
Tstat8 uses T3000 software to communicate with others.The T3000 Building Automation System is a complete solution which allows a building operator to manage the indoor environment of commercial, institutional and residential buildings. Equipment such as chillers, air handlers, boilers, fans, lights, pumps, dampers, valves and other mechanical equipment can be scheduled and controlled according to the sequence defined by the control system designers. Extensive monitoring and logging features allow the operator to see what is going on in their building to proactively improve comfort and efficiency. The system has powerful communications capabilities making it possible to integrate with other systems and can scale to virtually any size from a home sized system on up to a university campus with many buildings.



An operator with the appropriate privileges may access the entire system regardless of whether they are logged on over the internet or locally from a PC inside the building. End users and tenants can be set up with access to only their own areas which gives them the power to control their own area as they like.

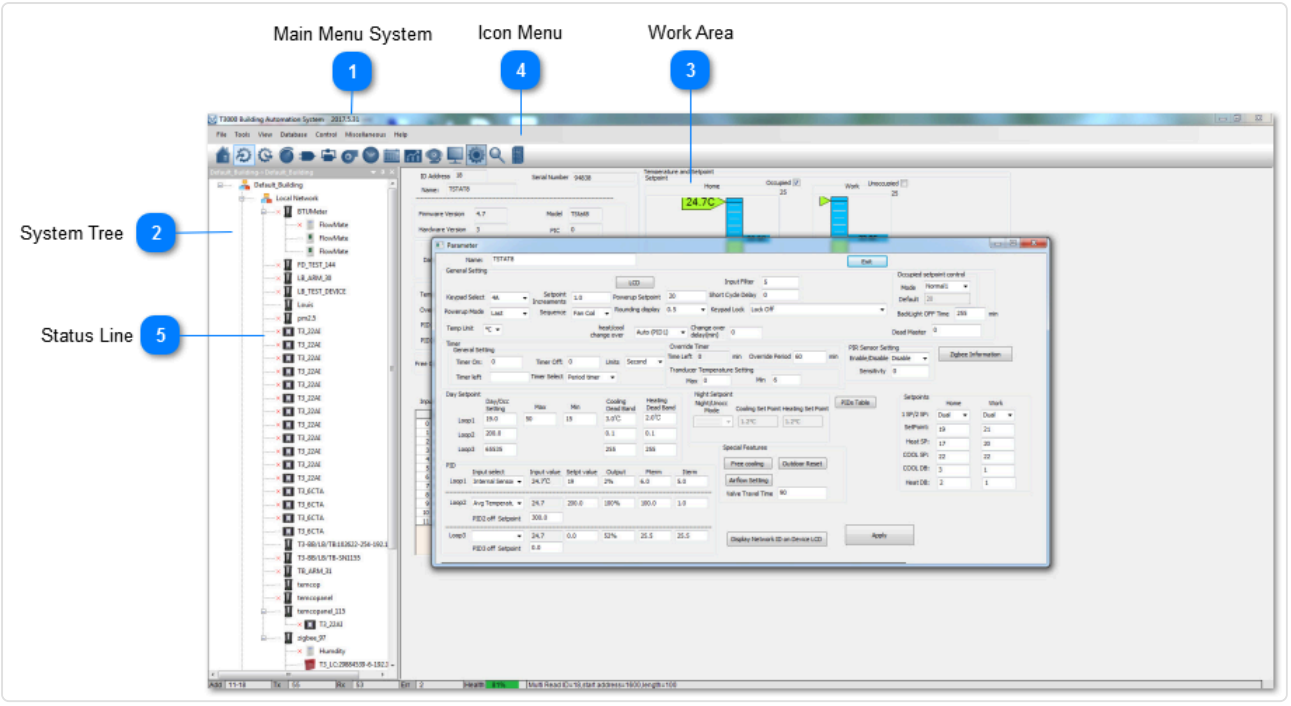
Simultaneous users may access the system over one of the many communications channels. Alarms, system graphics and trend data are stored directly on the controller so there is no central server or single point of failure.

Interactive graphic displays, floor plans and custom menus make operation simple for any user. The flexible T3000 hardware and software makes it easy to configure and reconfigure the system throughout the life of the building. The system is primarily targeted at HVAC applications but can also be used for many other applications such as lighting, green houses and other light industrial systems. The system also supports industry the standard bacnet and modbus protocols which means its possible to integrate with access control systems, lighting, power meters and others to form a complete building management system.



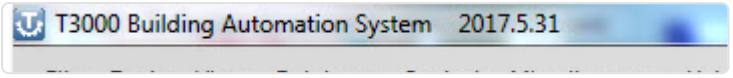
The contents of T3000 Software

Menus and Work Area



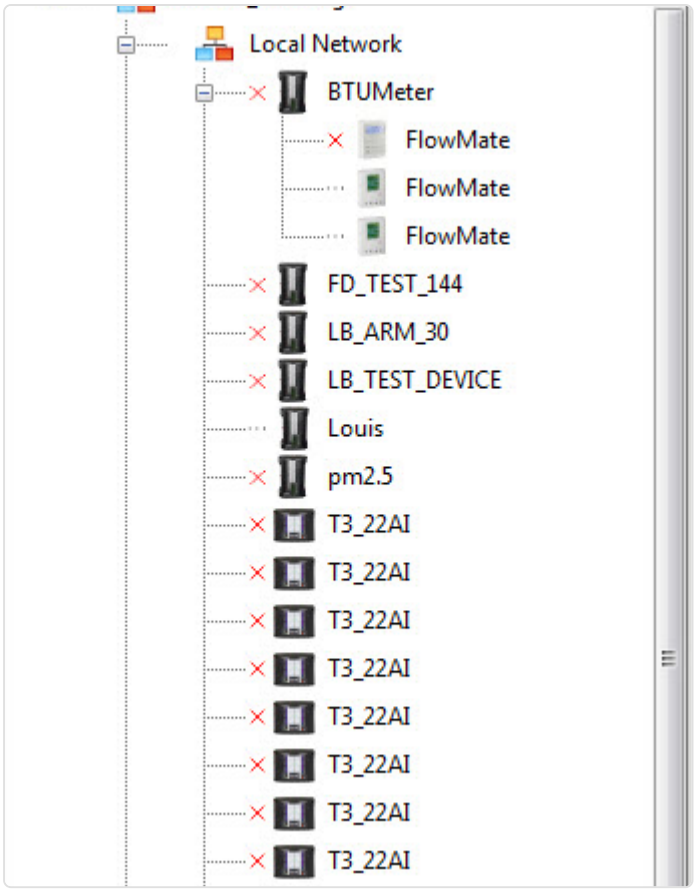
Tstat8 uses T3000 software to communicate with others.This is the Menu and Work Area of T3000 Building Automation System,including Main Menu System,Icon Menu,Work Area,System Tree,Status Line.

Main Menu System



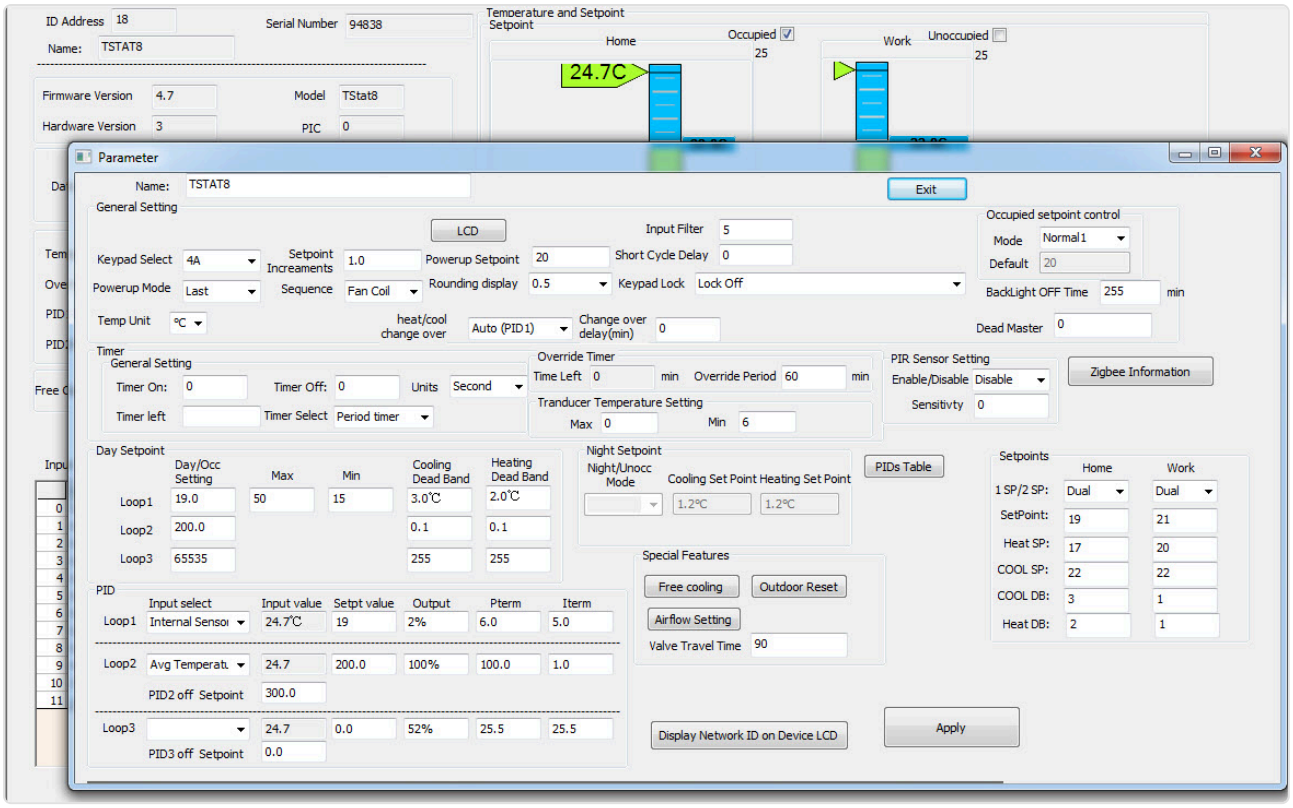
The main Menu system shows along the top of the screen.

System Tree



The System Tree section shows in this window area along the left hand panel. The tree can be turned off and sorted & edited in different ways. Items which are offline will have a small red 'X' near the node. The tree is described in more detail here in the System Tree section.

Work Area



This is the main work area which can be text based for working with grids and setup screens, or it can be graphical for working with the operator displays and floorplans. Setting up graphical displays is described in the Graphical Screens section

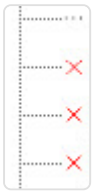
Icon Menu



Shortcuts to the most commonly used work areas of the system. Inputs, Outputs, User screens, trend logs and so on all presented here in icon format. There are also these same items available from the text menu system.

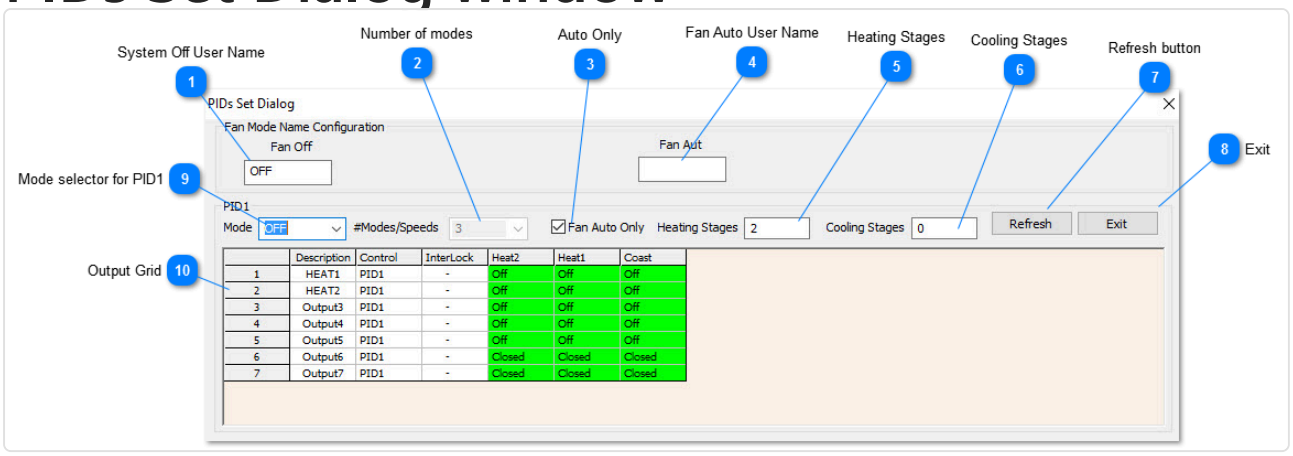
Hover text will show the function of each of these icons when you place the mouse over the icons without pressing any buttons.

Status Line

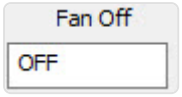


The communications status line shows the address of the device which is currently communicating in the Add box. The RX and TX numbers show how many packets have been sent to the current device in the current session, each time you click on a different device the numbers will reset to zero. The ERR box shows the number of errors in the current session with the current device. The Health box shows the percentage of successful packets for the current session for up to 100 packets. When the session hits more than 100 packets the earlier packets are pushed out of the tally, only the latest 100 are used in the math. Finally the status line shows some brief information of the data which is being sent and the status.

PIDs Set Dialog window

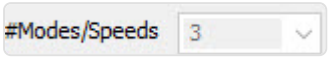


System Off User Name



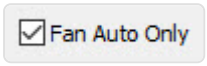
User name for when the system is off, can change this field to show "SHUTDOWN" , "UNOCCUPIED" or any other user defined name.

Number of modes



Define the number of modes for the system. Most systems will have at least two for the ON and OFF modes. Other systems like a fan coil with a multi speed fan would have five: OFF-LOW-MED-HI-AUTO. The minimum number of modes is fixed as 3 for ON-OFF and AUTO but you can add more to suit the application.

Auto Only



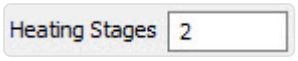
Check this box if you don't want to let the user step into the manual modes, the system will take care of staging the system from stage 1 cool to stage 2 cool and so on automatically. If you want the user to have more control at the keypad to let them put the system on manually, to move some air in the room for example by setting the fan on manually, leave this box unchecked. Note that in this case, the system may not run at its optimum efficiency, the fan would be running full time even when the setpoint is achieved.

Fan Auto User Name



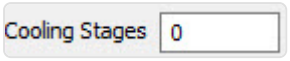
This is the user defined name for when the system is in the auto mode. It could be changed to RUNNING or OCCUPIED or whatever suits the application.

Heating Stages



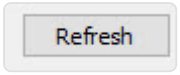
This is the number of stages of heating in the system. A single stage heater would set this item to 1. A two stage air conditioned would set this to two and so on.

Cooling Stages



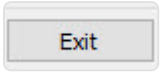
This is the number of stages of cooling in the system. For a single stage air conditioner set this to 1. For an air handler with two stages of cooling, set this to two and so on.

Refresh button



Self explanatory, hit this button to refresh the screen.

Exit



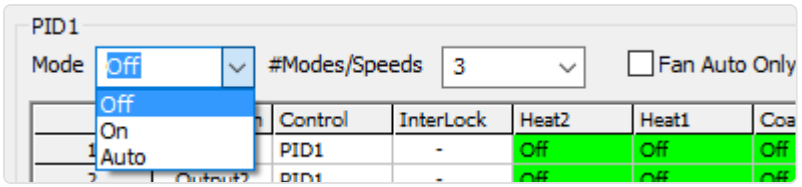
Saves the current changes and leaves the setup screen

Mode selector for PID1

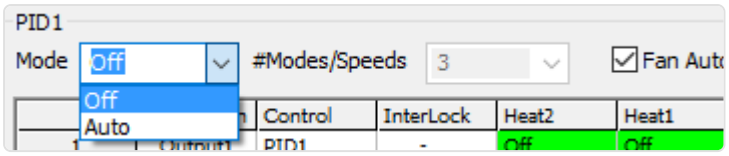


There are several tables to define the staging of the outputs in each mode of operation. Once the number of modes & speeds has been set this tab lets you view the output grid for each of the modes. Note that you can change the names of these modes so your screen may look a little different than described here.

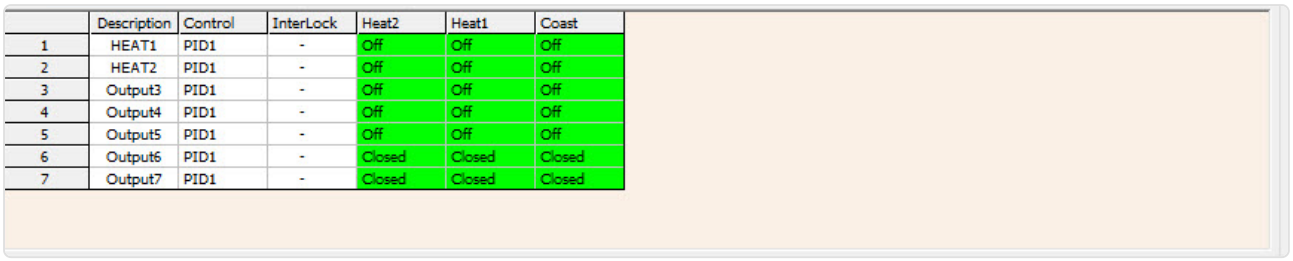
If you have 3 modes of operation, this tab will have the OFF-ON-AUTO.



However, if you set the Fan Auto Only box, the ON mode normally reserved for the manual ON settings will be disabled and the manual ON table option is no longer visible.



Output Grid



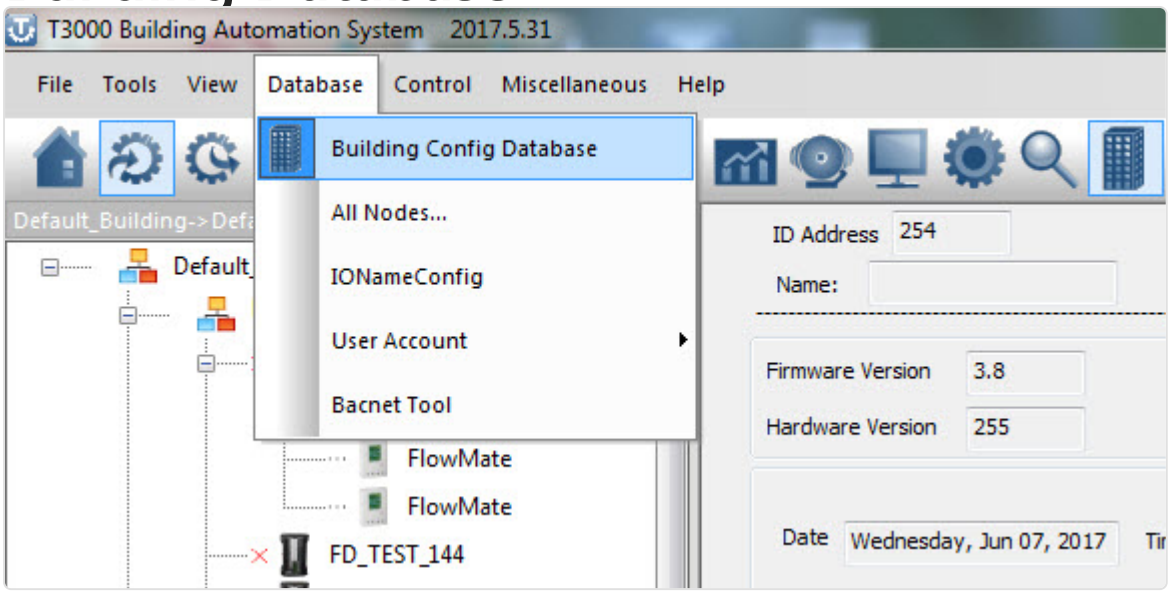
	Description	Control	InterLock	Heat2	Heat1	Coast
1	HEAT1	PID1	-	Off	Off	Off
2	HEAT2	PID1	-	Off	Off	Off
3	Output3	PID1	-	Off	Off	Off
4	Output4	PID1	-	Off	Off	Off
5	Output5	PID1	-	Off	Off	Off
6	Output6	PID1	-	Closed	Closed	Closed
7	Output7	PID1	-	Closed	Closed	Closed

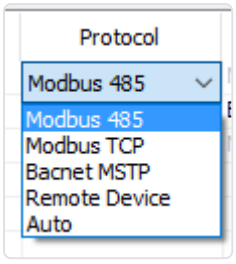
Here is where you define which output will be active for a given stage of heating or cooling. Each row represents one of the outputs and each column represents a certain stage of heating or cooling.

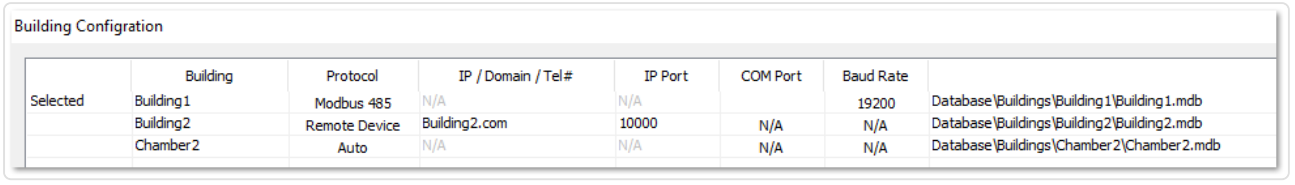
There are one or more 'Mode' tables, each of which is represented by one of these grids with on and off settings for each output over each stage of operation. Normally the occupied mode and the unoccupied mode are enough for a typical HVAC application but you are able to define more modes, such as low-medium-hi in the case of a three speed fan coil system. In the example above, the OFF mode is currently selected so the grid shows the output states when the system is operating in the OFF mode.

There are two collumns called heat1 and heat2 for the two stages of heating and no columns for cooling since cooling is set to zero. The Coast column represents the state of the outputs when the setpoint is satisfied and the system is said to be 'coasting'. In this example, all the cells are OFF so the system will not try to maintain the setpoint in this mode, all outputs are truly OFF. In cold weather you may want to maintain a certain unoccupied setpoint to keep the pipes from freezing in the building in which case the stage 2 heating would have some of the outputs flipped to ON.

Building Database







<Building Configuration

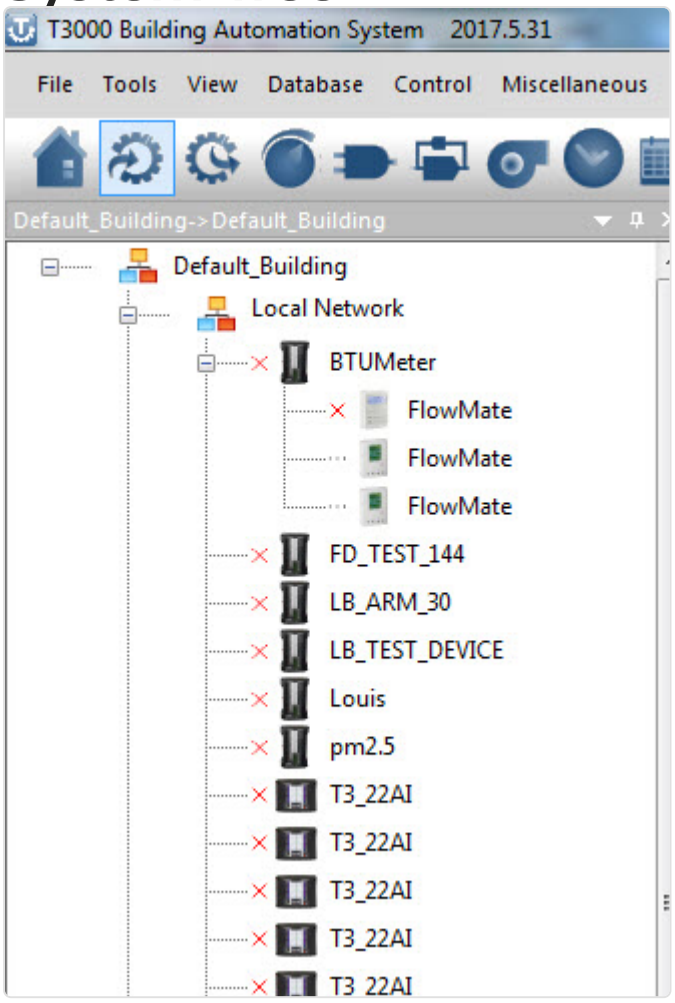
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notes: When protocol is auto ,IP ,Port ,Com Port ,Baud Rate disable.

When protocol is Modbus Tcp or Romote Device ,Ip,Port enable ,Comport Baud Rate disable.

When protocol is Modbus 485 or Bacnet MSTP ,IP,Port disable ,Comport Baud Rate enable.

System Tree



A controller with an RS485 subnetwork will show the subnet devices on a branch below the controller.

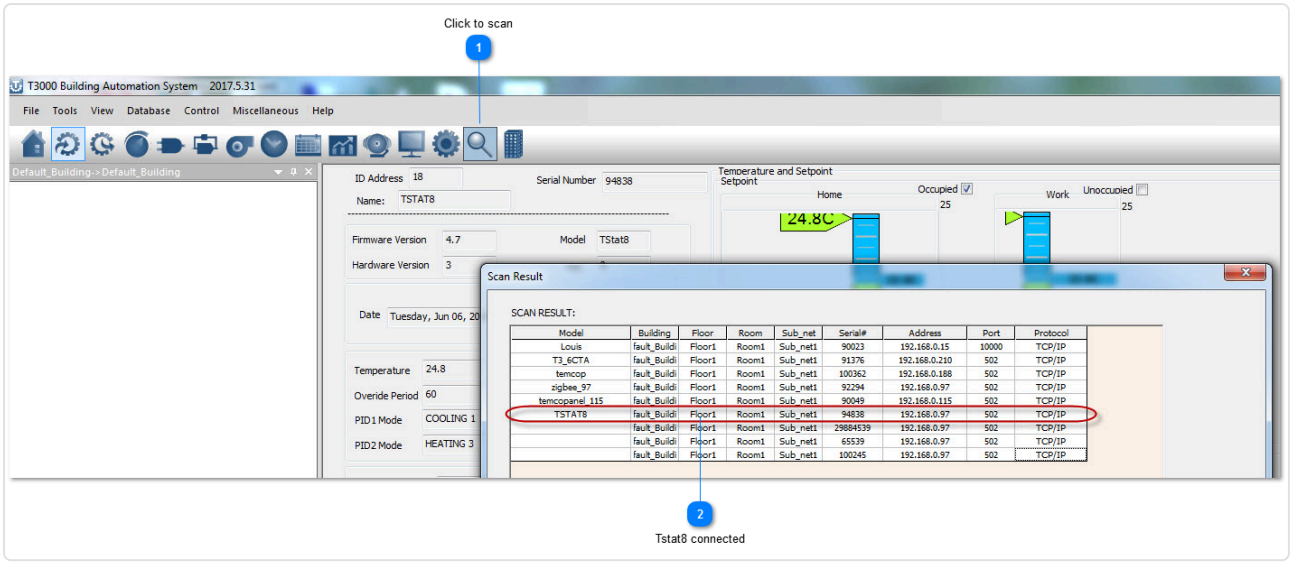
T3000 Operation

Step1.Visit 107.170.34.189/ftp/software/9TstatSoftware.zip, download T3000 software and install it;

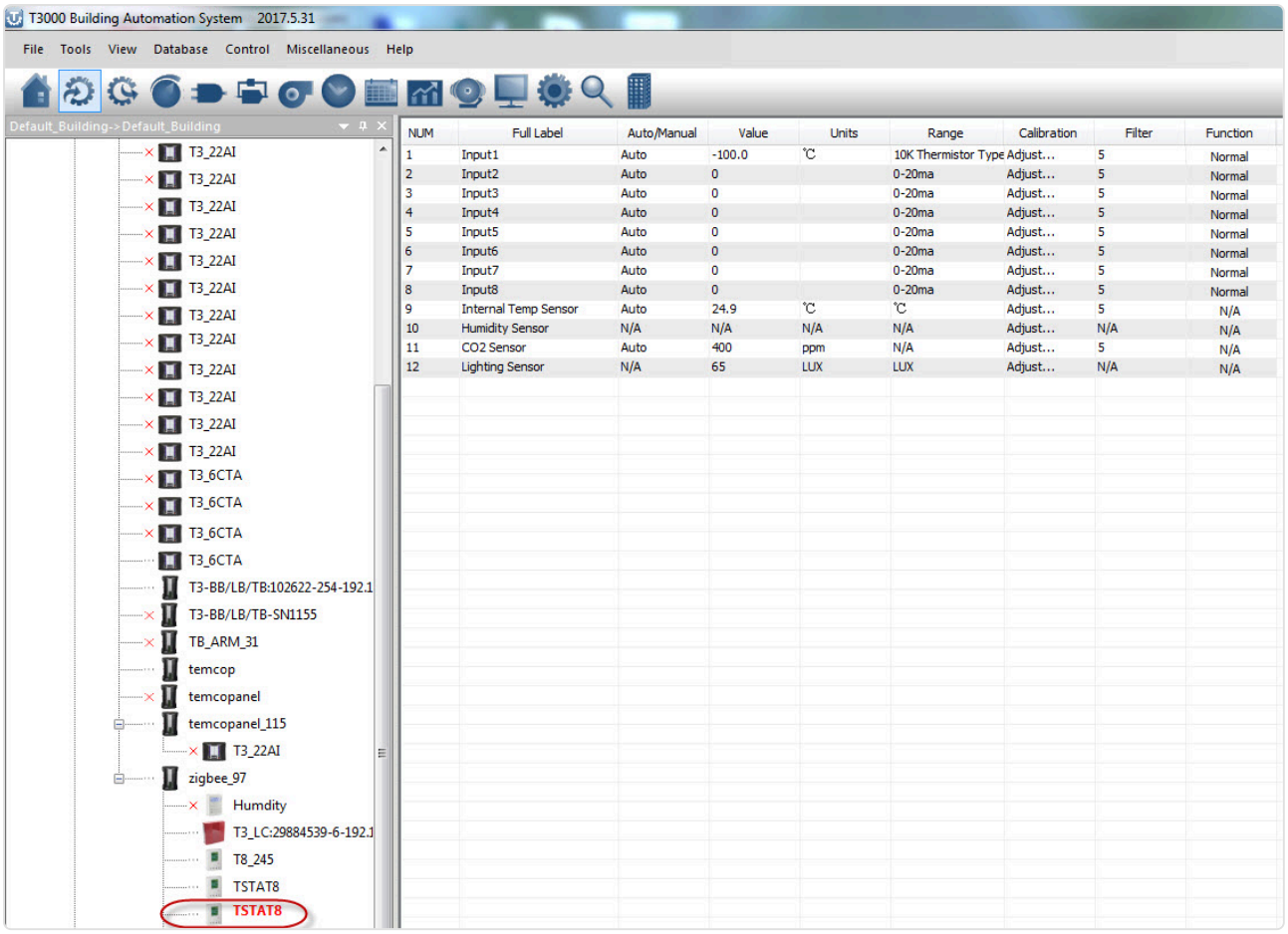
Step2.Plug Tstat8 in power,connect Tstat8 to PC via RS485 or Ethernet;

Step3.Start T3000 software,click  to scan,then you can find Tstat8 as below.

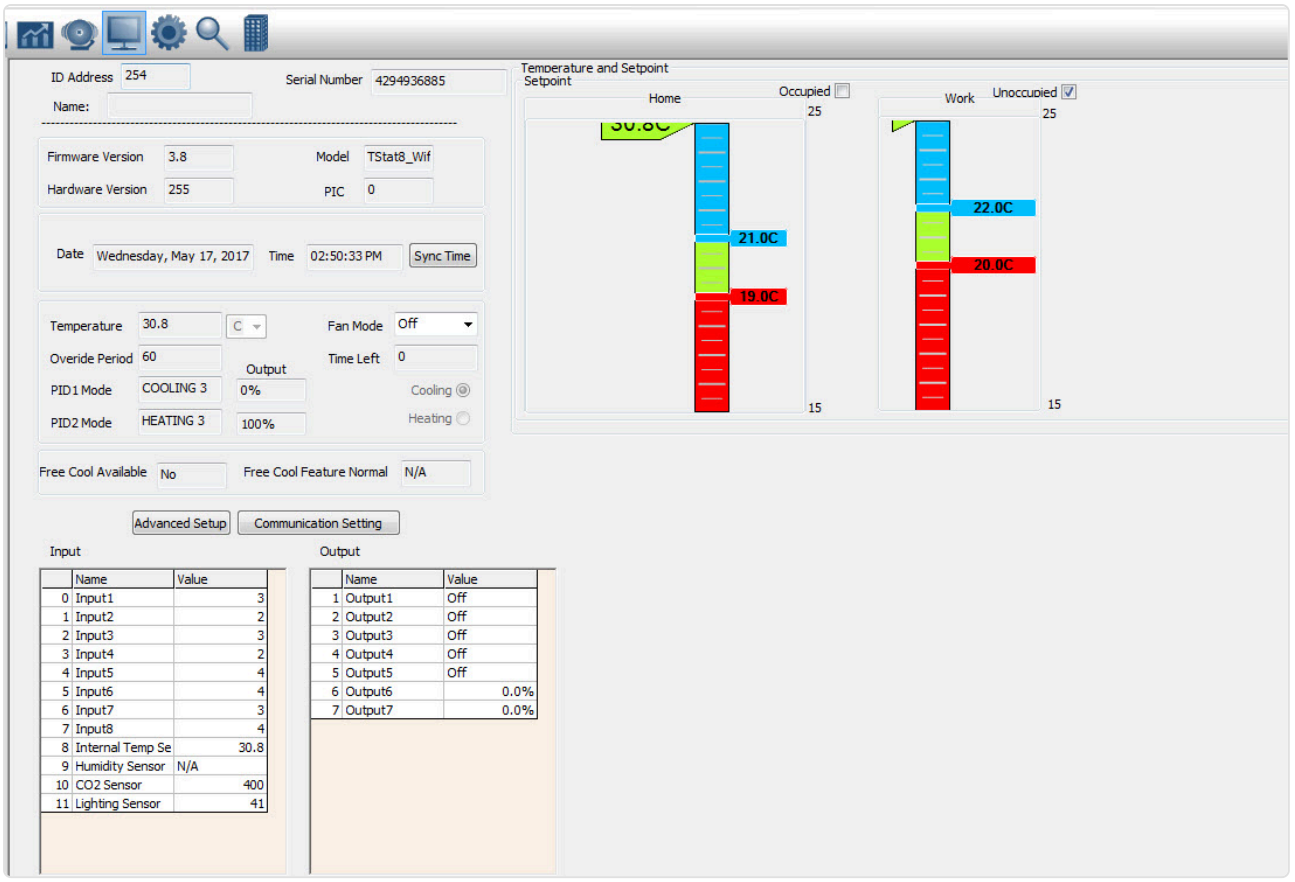




4. Click Tstat8 log, then click  , this will display all of the Tstat8 inputs. Change the name of the input and range that fis the application.



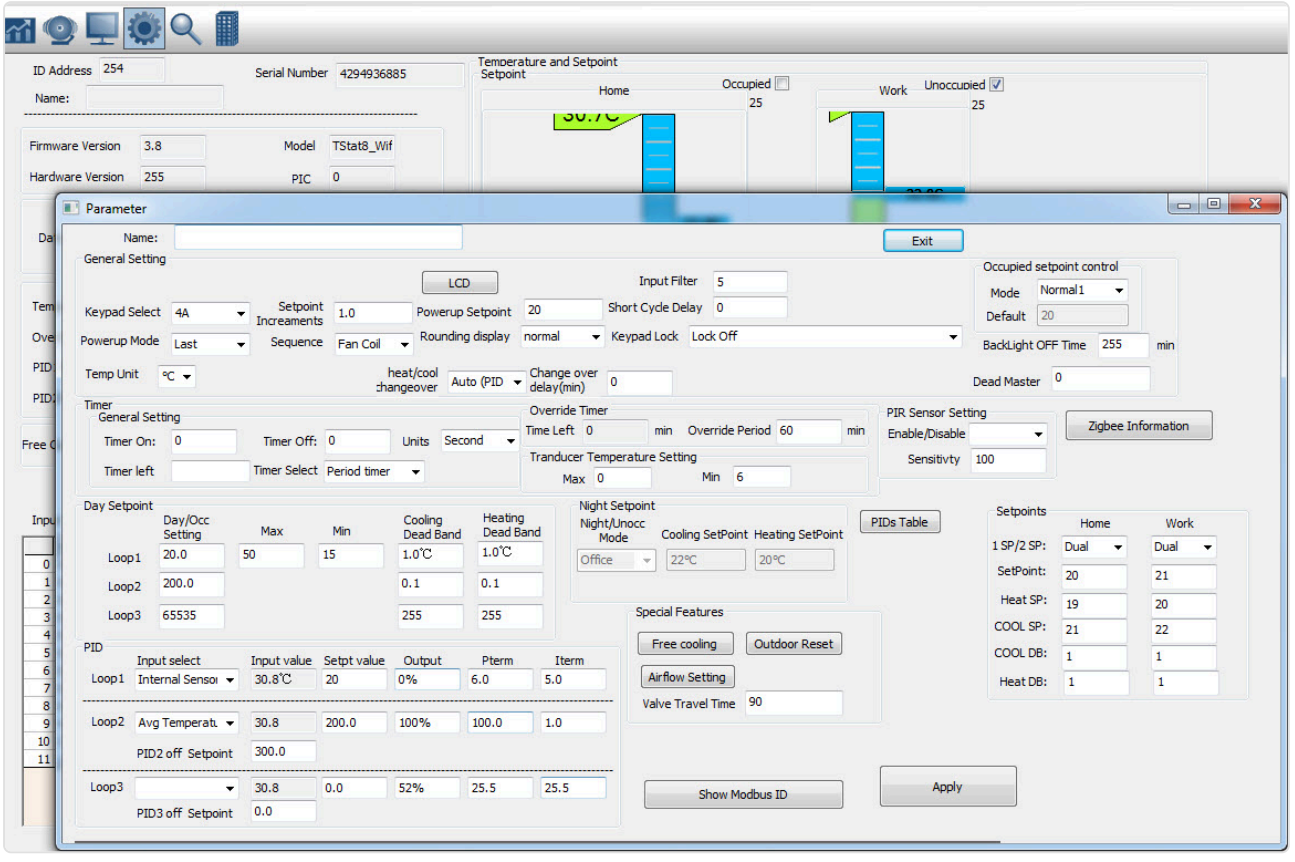
5.Click  to see the status of the Tstat8. This window will display setpoints, temperature, inputs and outputs.

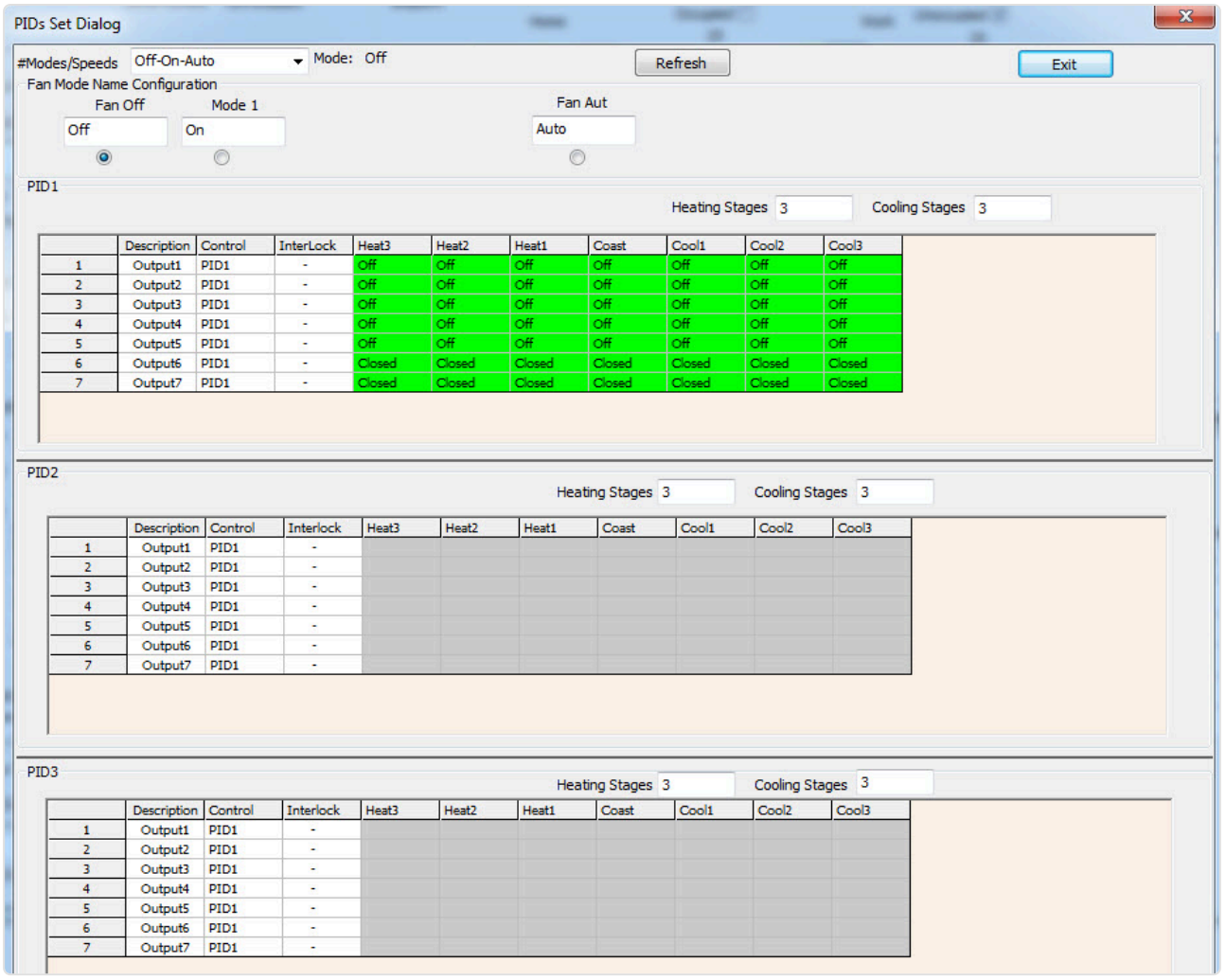


	Name	Value
0	Input1	3
1	Input2	2
2	Input3	3
3	Input4	2
4	Input5	4
5	Input6	4
6	Input7	3
7	Input8	4
8	Internal Temp Se	30.8
9	Humidity Sensor	N/A
10	CO2 Sensor	400
11	Lighting Sensor	41

	Name	Value
1	Output1	Off
2	Output2	Off
3	Output3	Off
4	Output4	Off
5	Output5	Off
6	Output6	0.0%
7	Output7	0.0%

6.Click  to show parameter tab below,you can do the settings.For example,click□PIDs Tables□to□PIDs set Dialog□,click to set □Heating Stages□value.





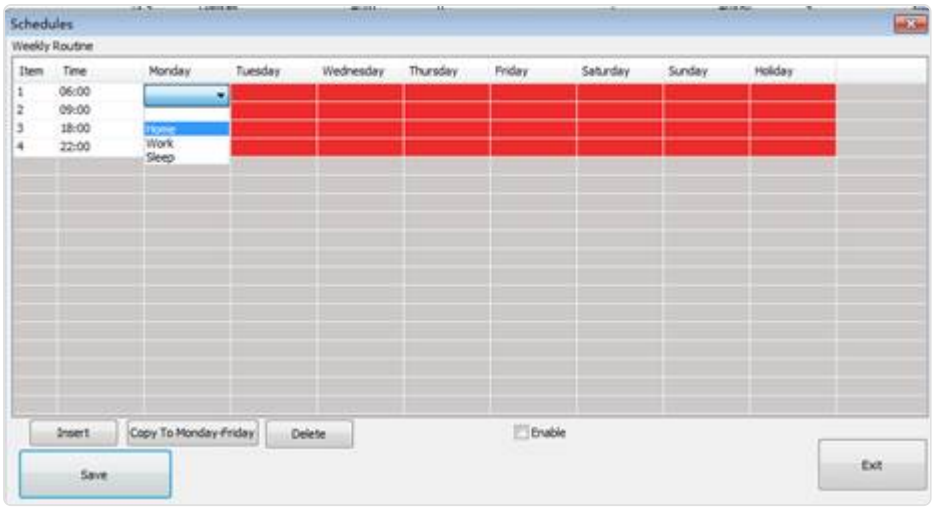
7.Click the schedule icon  to go to schedule window,you can do schedule settings.

Each day we support 6 events, and you can select the mode for each event:

For home mode, unit will use day setpoint to control the outputs;

For work mode, it will use night setpoint to control the outputs;

For sleep mode, it will use sleep setpoint to control the outputs



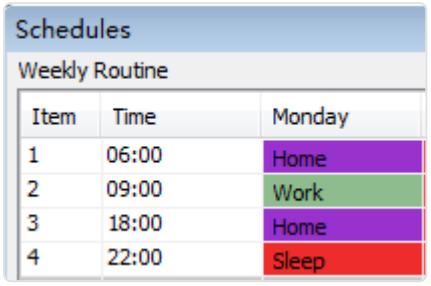
For example as below:

when time is between 6:00 to 9:00, unit will work on home mode;

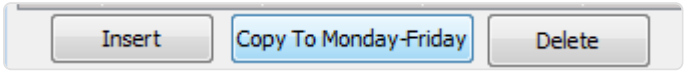
when time is between 9:00 to 18:00, unit will work on work mode;

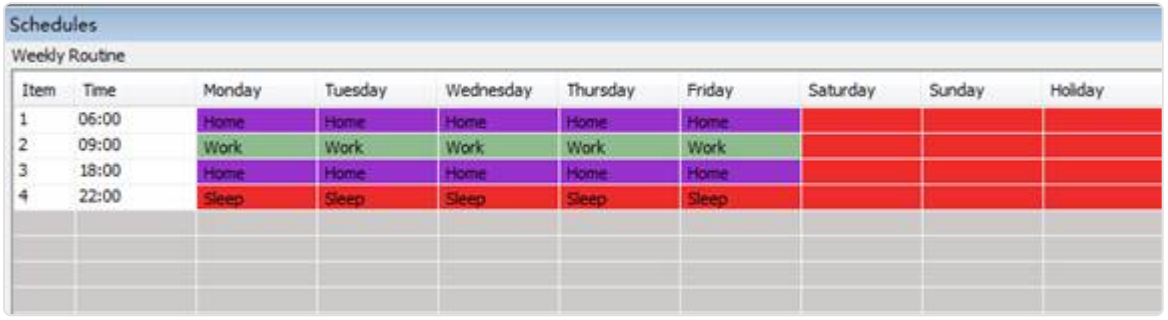
when time is between 18:00 to 22:00, unit will work on home mode;

when time is between 22:00 to Tuesday 6:00, unit will work on sleep mode



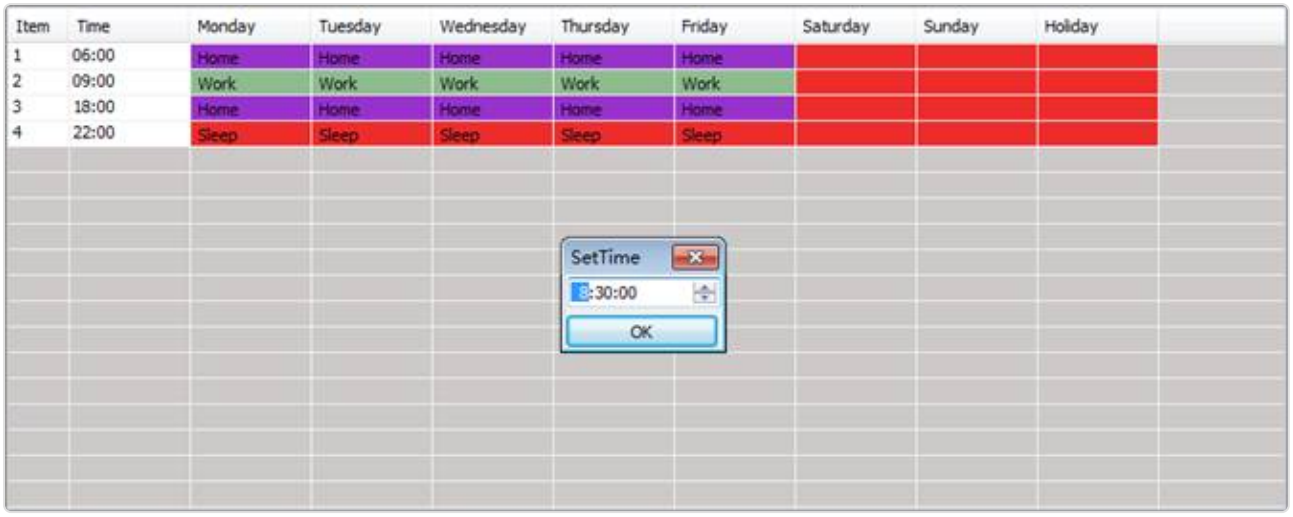
After you setup the Monday schedule, you can copy the Monday setting from Tuesday to Friday, then you can use the "Copy To Monday-Friday" function to copy the setting.

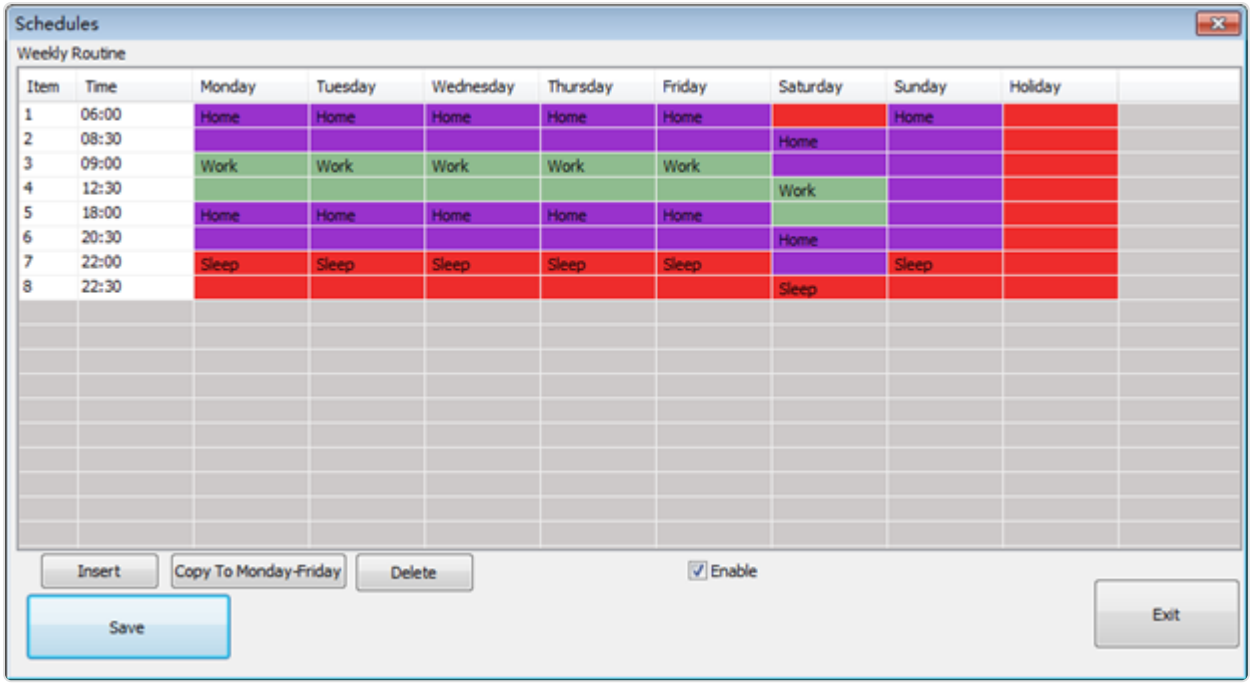




If you need different setting for each day, you can use insert function to edit your schedule, for example, on Saturday, double click the window and it will show a small dialog to insert new event

and you can set up the time.





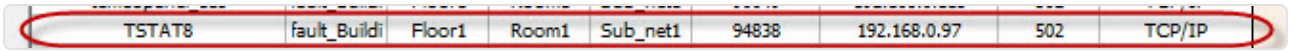
So on Saturday, when time is between 8:30 to 12:30, unit will work on home mode;when time is between 12:30 to 20:30, unit will work on work mode; when time is between 20:00 to 22:30, unit will work on home mode;when time is between 22:30 to Sunday 6:00, unit will work on sleep mode.

Note: Select □Enable□ option to enable schedule function.After edit schedule, make sure click the save button to save the setting !

Click to scan



Tstat8 connected



TSTAT8	fault_Buildi	Floor1	Room1	Sub_net1	94838	192.168.0.97	502	TCP/IP

Wireless Connections

The system supports an optional zigbee add-on module which enables connecting to wireless thermostats and other zigbee devices. The zigbee devices form a mesh which makes the network more robust than a regular point to point type network. If a particular node experiences interference or goes offline the network will heal itself and work around the problem node.



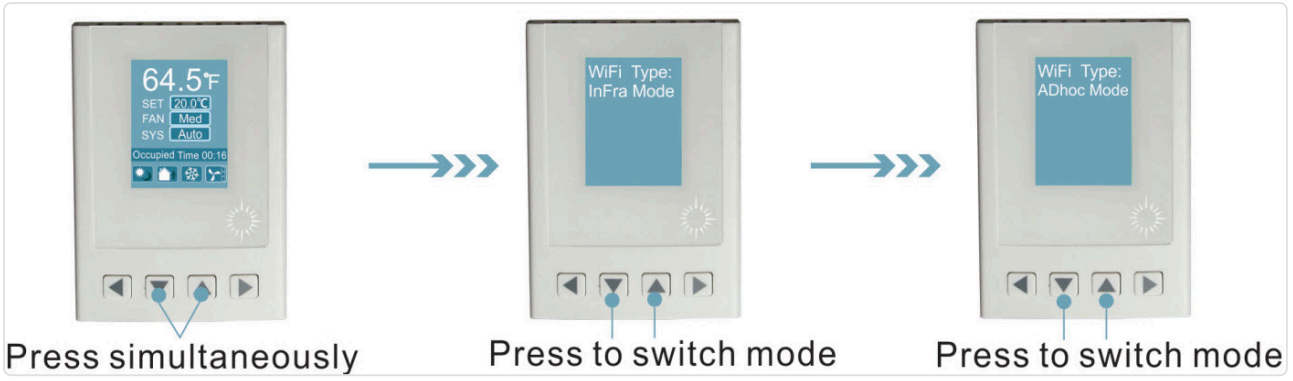
The contents of Wireless Connections

Wifi Configuration

1.Confiuration Settings

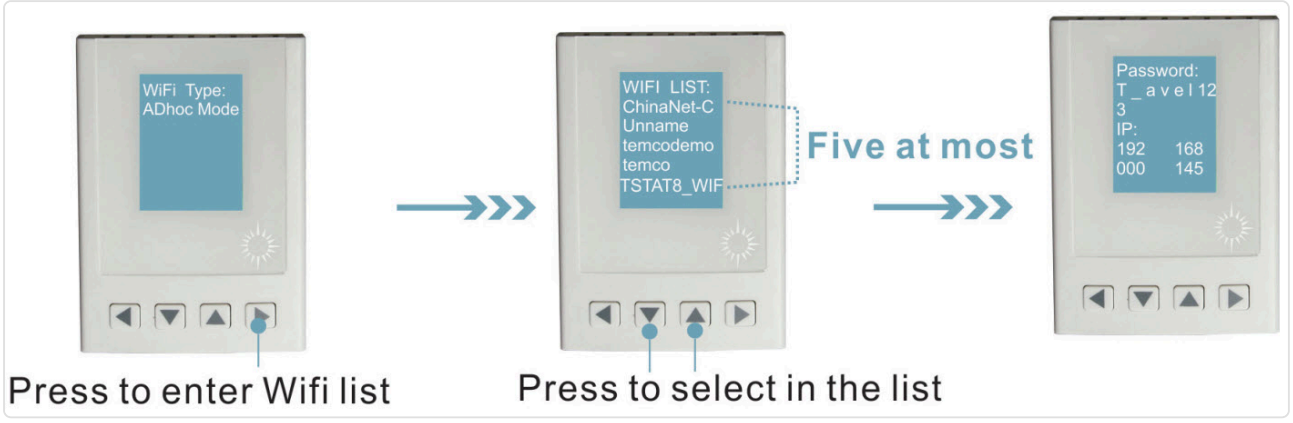
To set the password and IP address of the Tstat8, two methods are available:Key setting and ADhoc setting, or using the T3000 software Key setting.

Sep1. Apply power to the Tstat8.Press  simultaneously to enter mode selection.There are two modes:□Infra Mode□ and □ADhoc Mode□,press  or  to switch mode.



** **

Sep2.Press  to enter □WIFI LIST□ in mode selection.Press  or  to select WIFI in the list,press  or  to go up,and to go down.



Sep3.Press  to enter Password and IP setting menu.There are letters and numbers in the setting ,with a space character between them.The numbers are from 0 to 9,the capital letters from A to

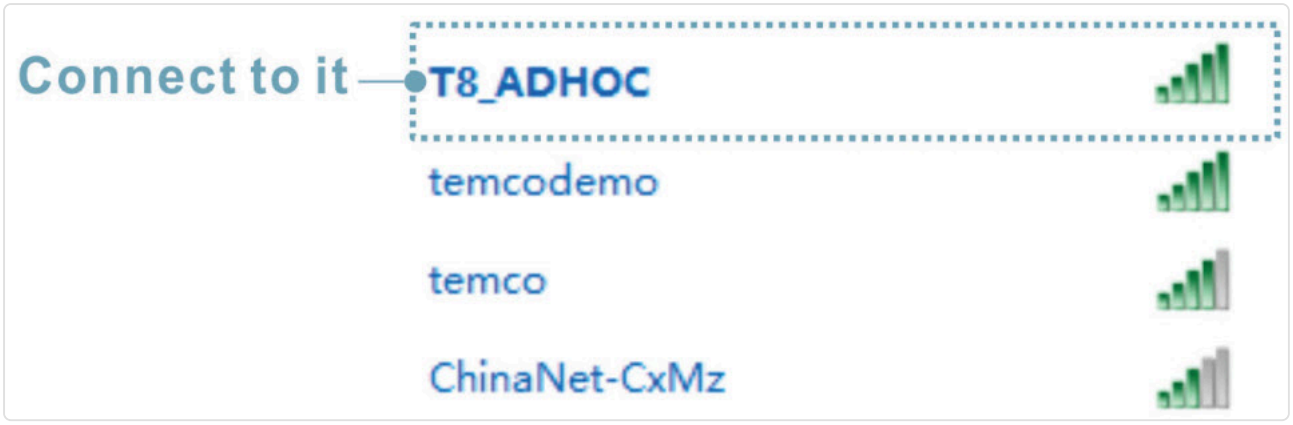
Z,and the small letters from a to z. Press  or  to select character,press  to go up,and  go down.

*It will return to the main menu if there isn't operation command in 3 seconds.

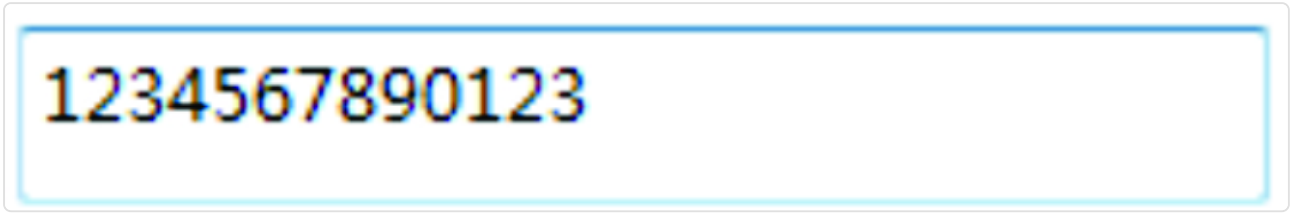
ADhoc setting

ADhoc is a singel hot model,which supports iphone,win7&8.The IP default address is 192.168.10.10.

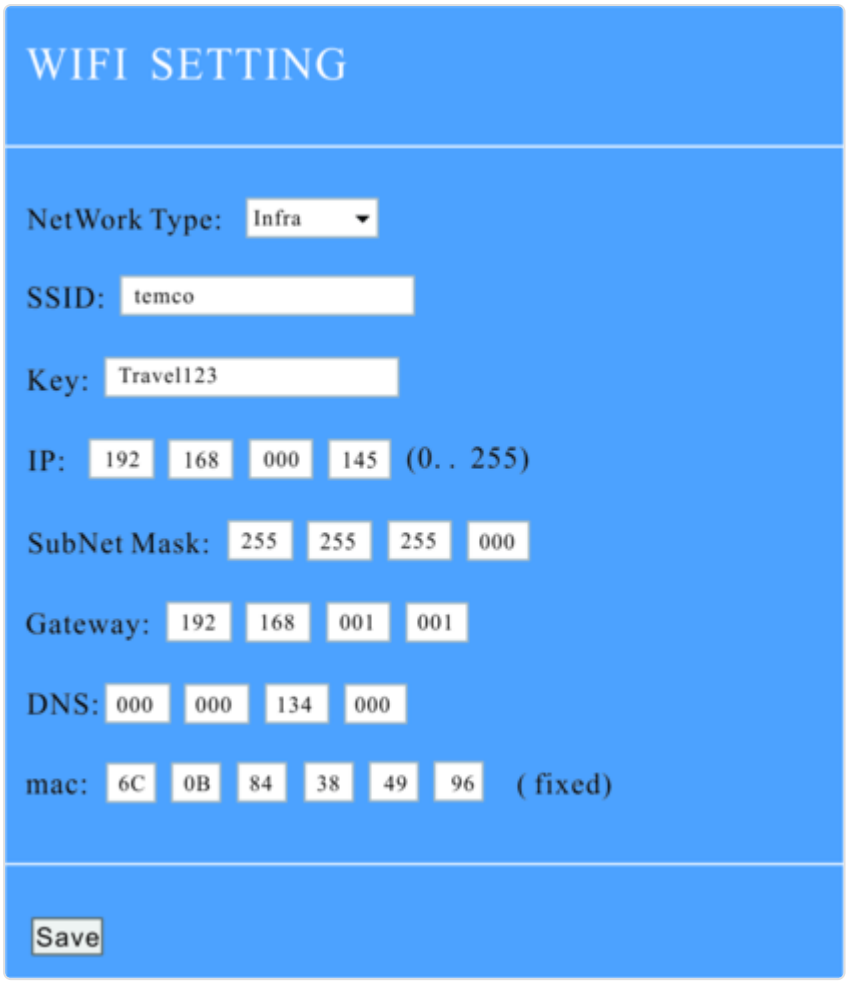
Sep1.To set passwords and IP addresses in Adhoc mode, start the Tstat8 fist, then scan for a wifisignal named "T8_ADHOC" from a computer with wifi connect to the wifi.



Step2.Enter the security key. The key of Tstat8 is always 1234567890123



Step3.Open your browser,enter IP 192.168.10.10,then you can set your password and IP as below:



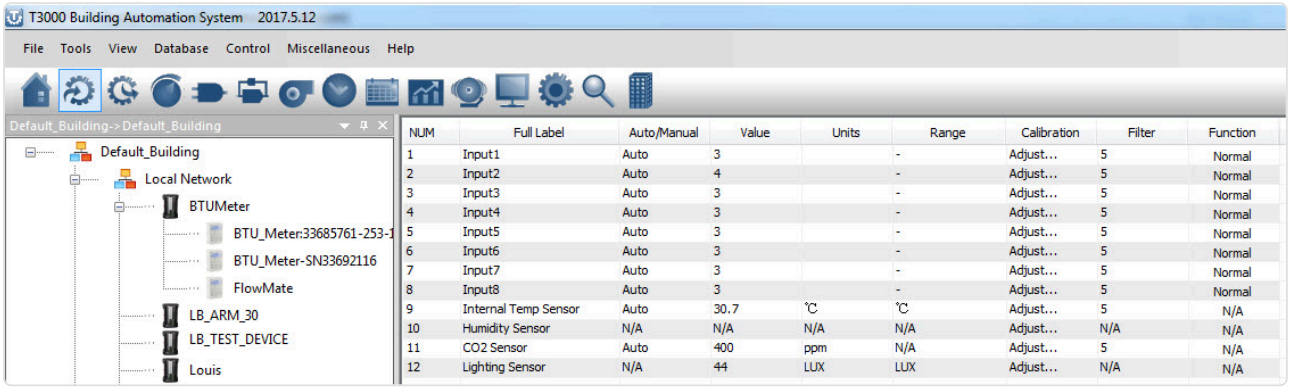
T3000 setting

The T3000 software also has the ability to set the password and IP of the Tstat8.

Step1.Visit https://temcocontrols.com/ftp/software/9TstatSoftware.zip, download T3000 software and install it;

Step2. Apply power to the Tstat8, connect the Tstat8 to a PC via RS485 or Ethernet;

Step3.Start T3000 software,click to scan,then you can fid Tstat8 as below.

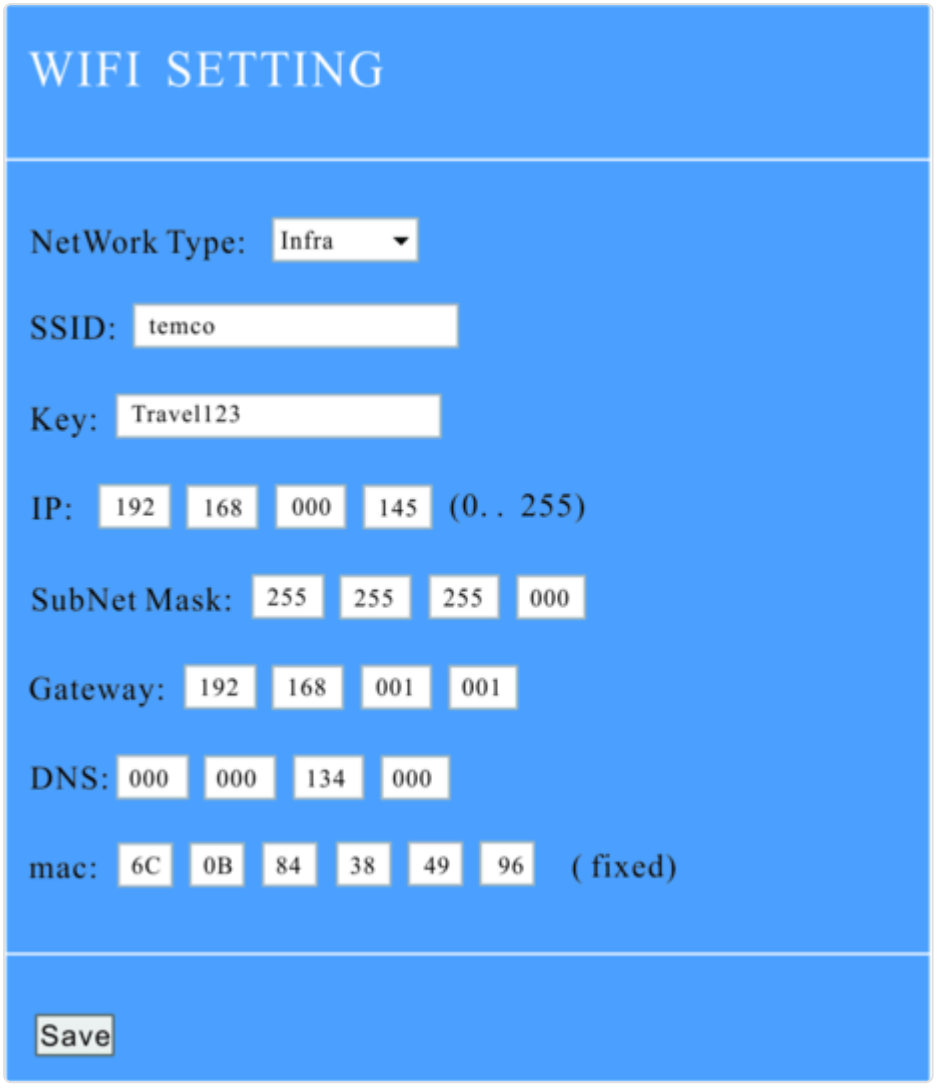


2.Application Settings

Infra mode is a typical mode in application settings. It works via a Ethernet router.

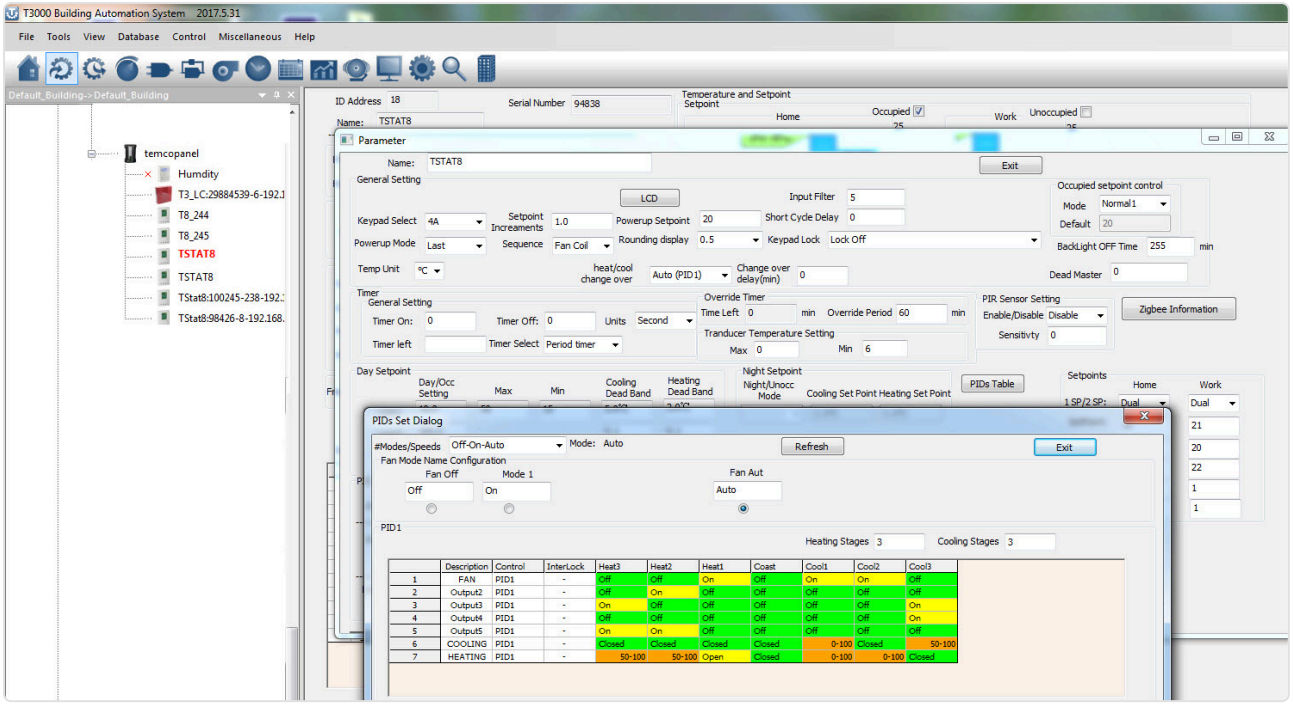
Step1.To set up in Infra mode, Apply Power to the Tstat8, connect your computer to the Ethernet router.

Step2.Open your browser, enter IP 192.168.0.145 and then you can set your password and IP Address.



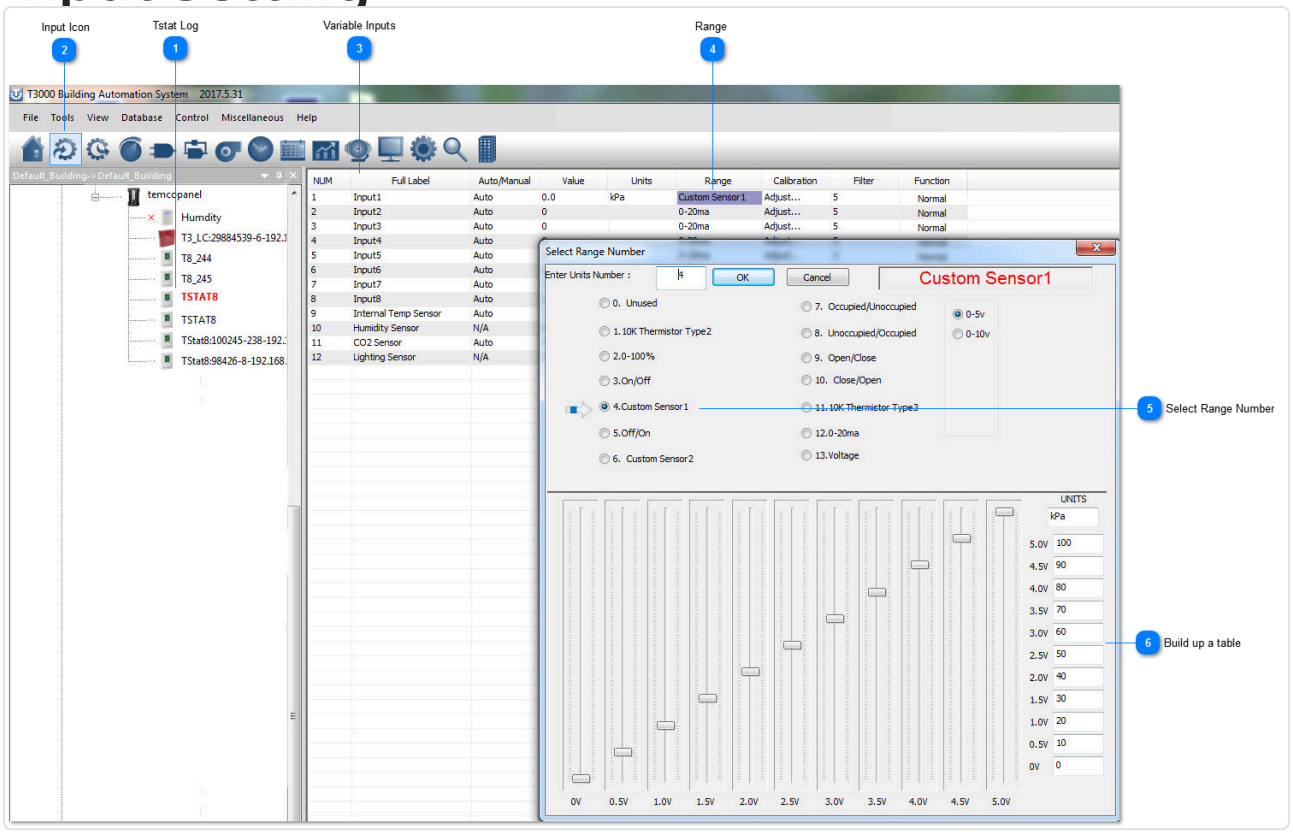
Heating/Cooling Configuration

About Heating Cooling Mode Configuration,here are two examples: one heat one cool setting and two heat two cool setting.



The contents of Heating/Cooling Configuration

Input Setting



As an example of a custom sensor, here we have built up a table for a custom sensor operating from 0 to 5V over the range of 0-100psi.

Tstat Log



Select the thermostat from the tree.

Input Icon



Click on the INPUTS icon.

Variable Inputs

NUM	Full Label
1	Input1
2	Input2
3	Input3
4	Input4
5	Input5
6	Input6
7	Input7
8	Input8
9	Internal Temp Sensor
10	Humidity Sensor
11	CO2 Sensor
12	Lighting Sensor

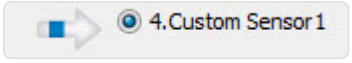
Give each input a name.

Range



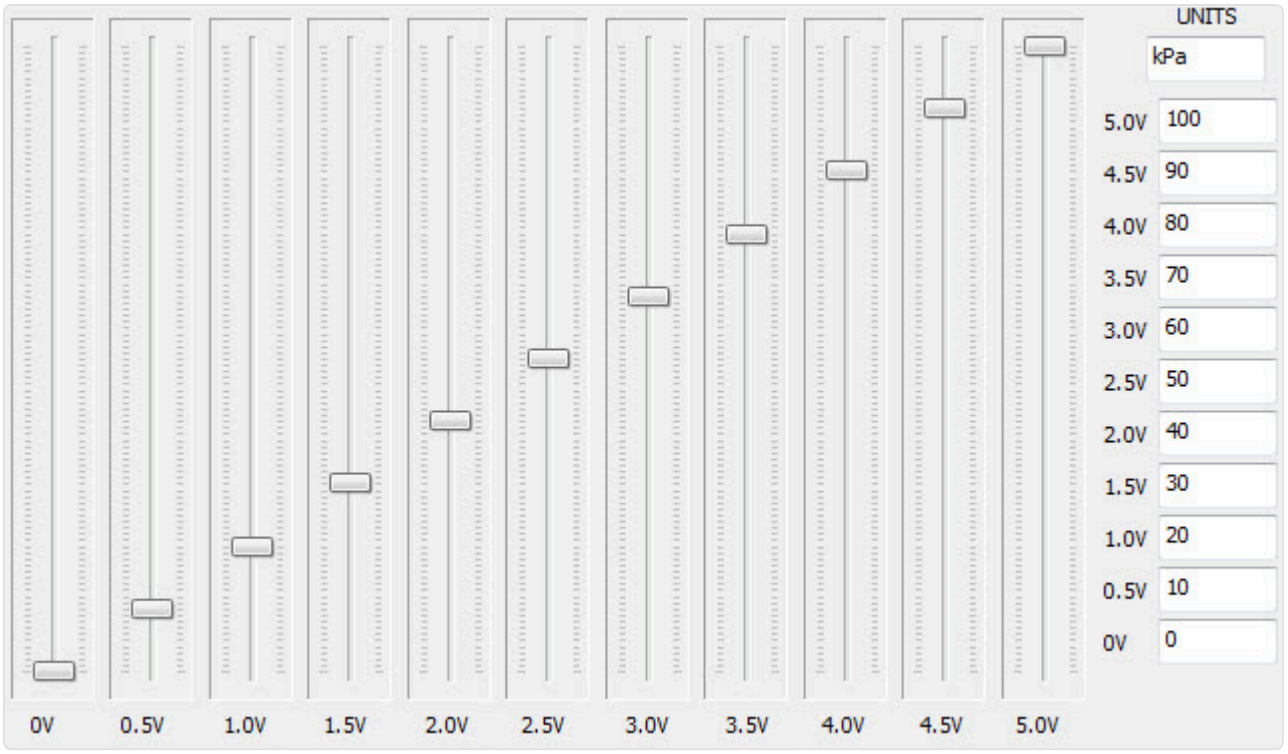
Set the range for the input by clicking on this column.

Select Range Number



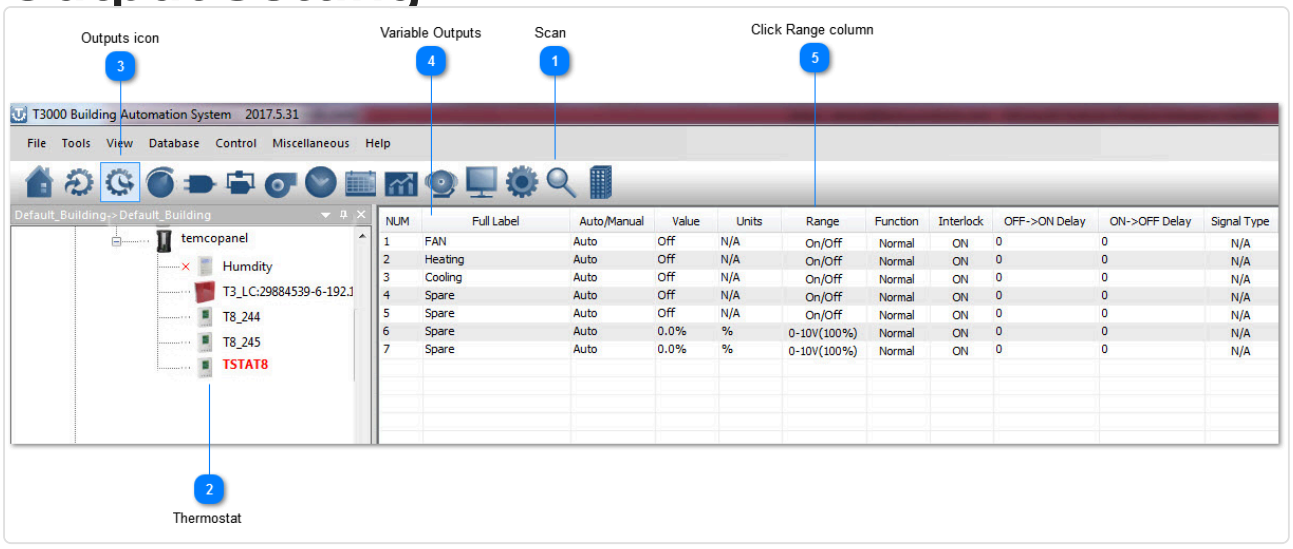
Select from the various ranges or build your own.

Build up a table



As an example of a custom sensor, here we have built up a table for a custom sensor operating from 0 to 5V over the range of 0-100psi.

Output Setting



Scan



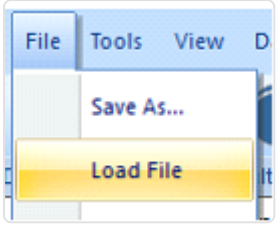
Scan the network and discover all devices.

Thermostat



Select the thermostat from the tree.

For a fast way to set up the stat you can just load the config file, attached. The steps I did to create this config file are explained below.



Outputs icon



Select the outputs icon.

Variable Outputs

NUM	Full Label
1	FAN
2	Heating
3	Cooling
4	Spare
5	Spare
6	Spare
7	Spare

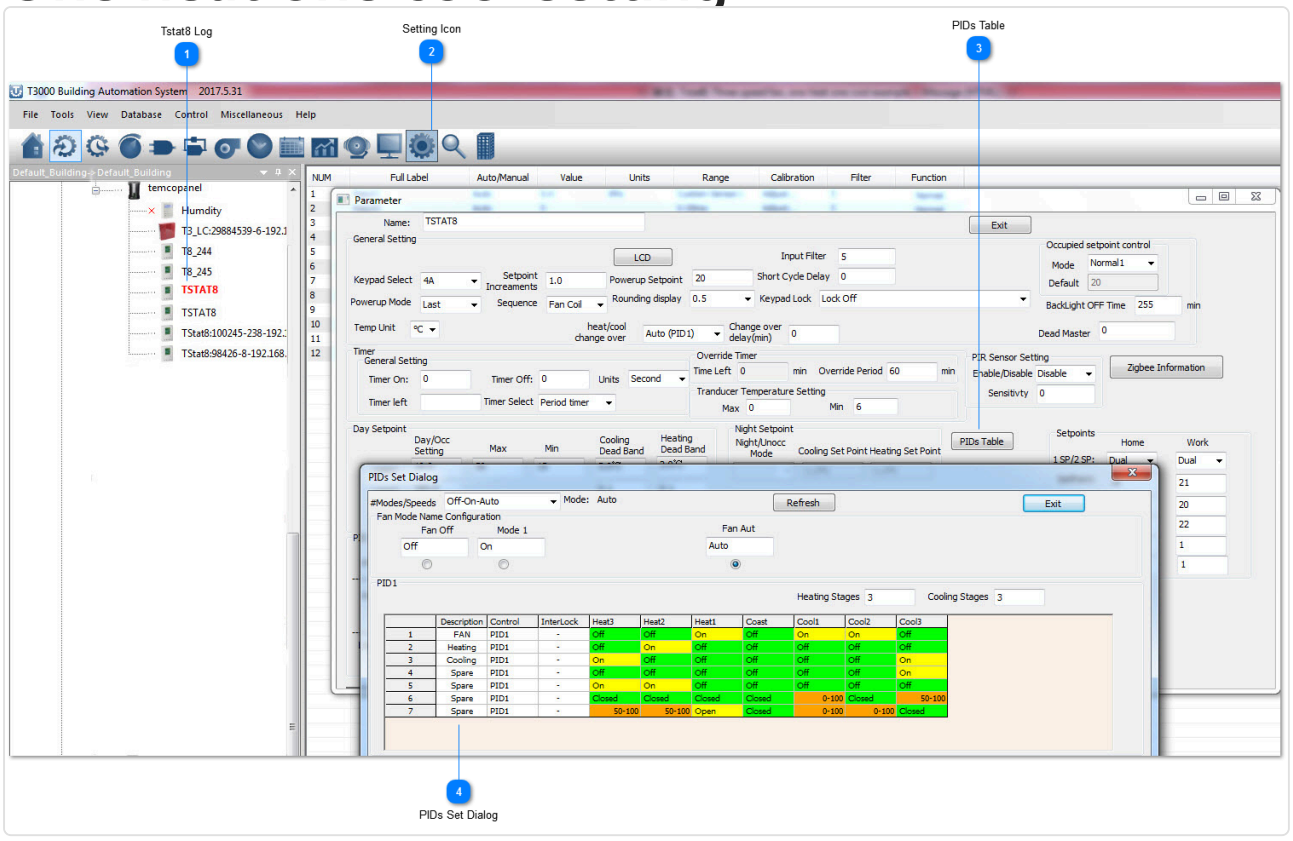
Click to give each output a name and a range, in the case of the fan the outputs are on-off and the valves are modulating 0-10V = 0-100% which are the default ranges already.

Click Range column

Range
On/Off
On/Off
On/Off
On/Off
On/Off
0-10V(100%)
0-10V(100%)

Click the Range column to see many options available for the range setting such as PWM and floating three wire control for modulating actuators using two relay outputs.

One heat one cool setting



Tstat8 Log



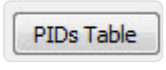
Select the thermostat from the tree.

Setting Icon



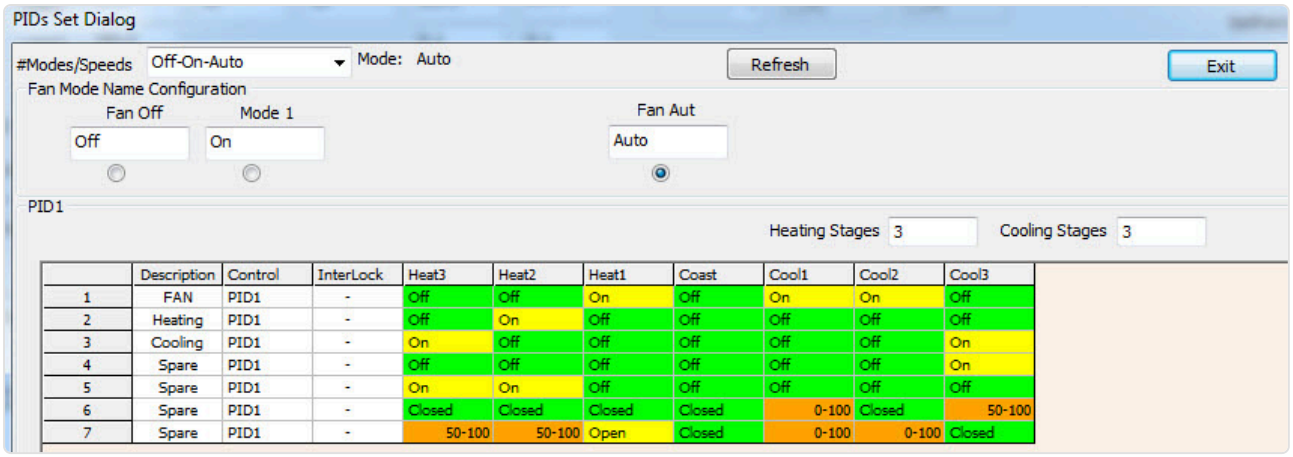
Click on the gear icon to get to the advanced settings.

PIDs Table



Click to get to the PID tables, this is where we assign outputs to act over each stage of heating and cooling.

PIDs Set Dialog

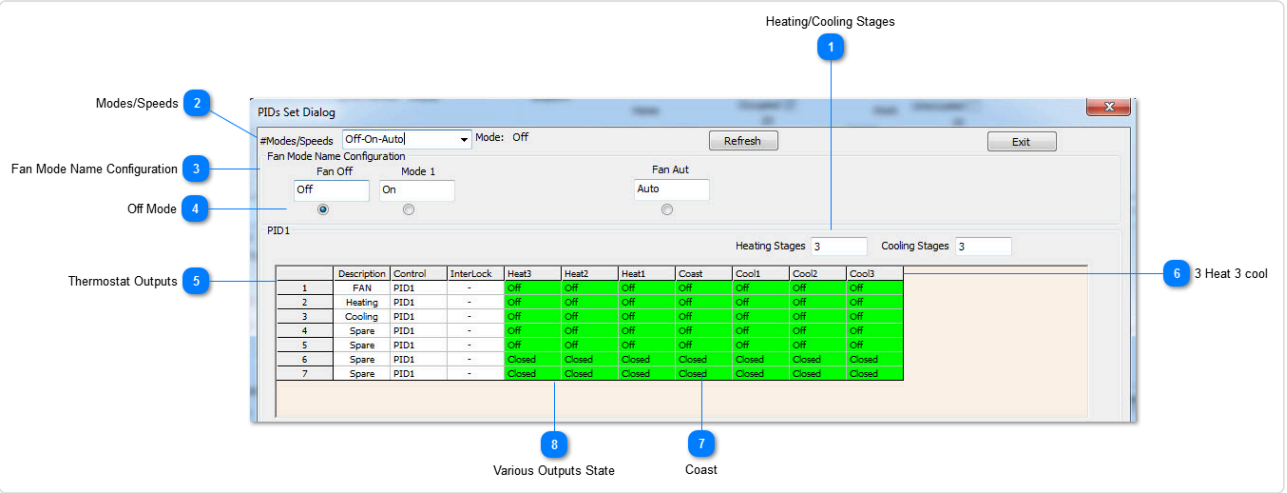


	Description	Control	InterLock	Heat3	Heat2	Heat1	Coast	Cool1	Cool2	Cool3
1	FAN	PID1	-	Off	Off	On	Off	On	On	Off
2	Heating	PID1	-	Off	On	Off	Off	Off	Off	Off
3	Cooling	PID1	-	On	Off	Off	Off	Off	Off	On
4	Spare	PID1	-	Off	Off	Off	Off	Off	Off	On
5	Spare	PID1	-	On	On	Off	Off	Off	Off	Off
6	Spare	PID1	-	Closed	Closed	Closed	Closed	0-100	Closed	50-100
7	Spare	PID1	-	50-100	50-100	Open	Closed	0-100	0-100	Closed

PIDs Set Dialog shows the details of the setting.

The contents of One heat one cool setting

OFF Mode

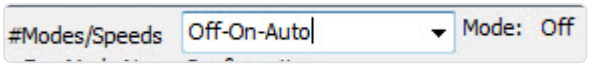


Heating/Cooling Stages



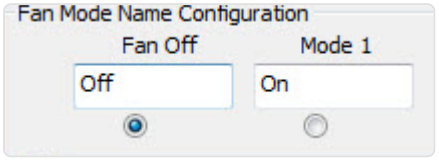
Here we set how many stages of heating and cooling the system will have. Since this is a three speed fan we can set three heating/cooling at this tab.If we set 2 or other number of heating and cooling stages,there will be corresponding quantity of tabs.

Modes/Speeds



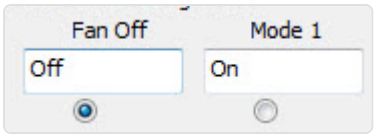
Select here from the available options, this establishes the various modes the thermostat will operate in and also whether the user will be able to set the stat in certain modes or not. For example if you select OFF-ON-AUTO the user will be able to select up to three modes from the keypad: OFF, ON and AUTO mode. If we had selected only OFF-AUTO, the user will only be able to see select from the OFF and AUTO modes. Keep in mind that the keypad can be locked as well, this is a separate setting, but this is where we set the number of modes the system will operate in.

Fan Mode Name Configuration



Each of these modes we established in the tab at 2 can be renamed along the row here at tab 3.

Off Mode



Now we set up which of the outputs will do what in each of the various stages and modes. We have selected the OFF mode and the state of the outputs is for the off mode.

Thermostat Outputs

	Description
1	FAN
2	Heating
3	Cooling
4	Spare
5	Spare
6	Spare
7	Spare

Each row represents one of the thermostat outputs.

3 Heat 3 cool

Heat3	Heat2	Heat1	Coast	Cool1	Cool2	Cool3

Each column represents a certain stage of heating or cooling. Heating is to the left and Cooling is to the right. The columns to the left represent increasing responses to the temperature being below setpoint. Moving to the right are increasing responses to the room temperature being over setpoint.

Coast

Coast
Off
Off
Off
Off
Off
Closed
Closed

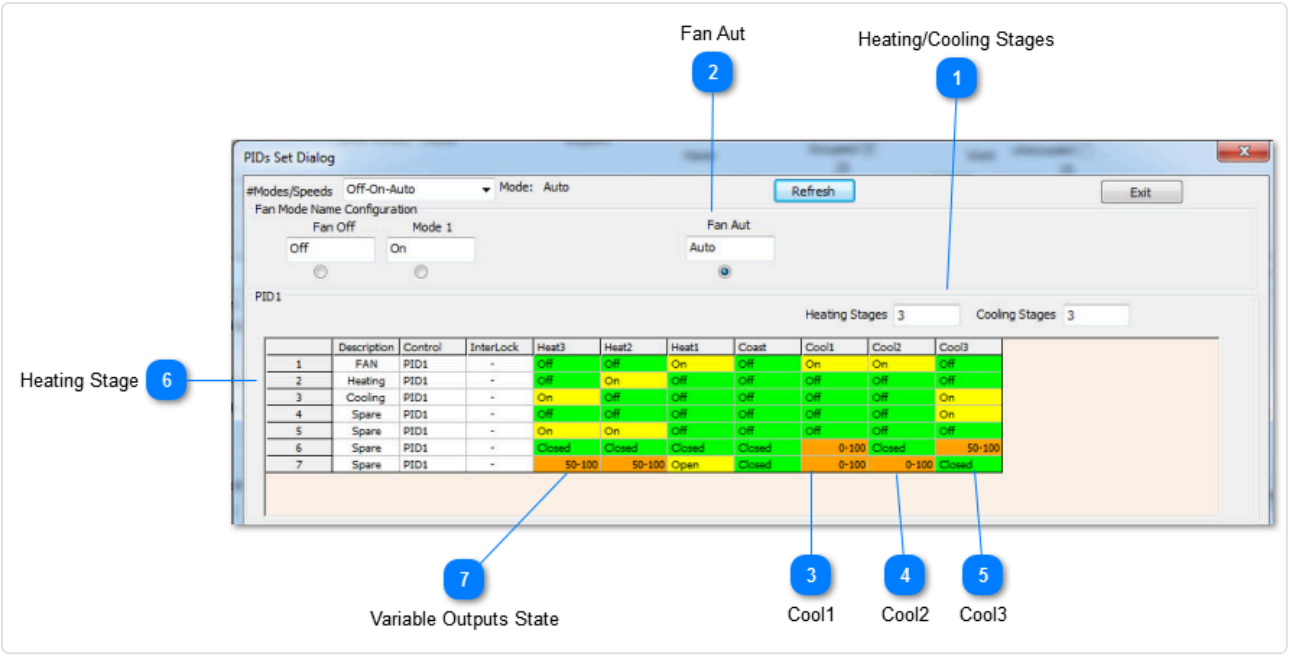
The center column represents the thermostat at rest, the setpoint is satisfied and the system is coasting.

Various Outputs State

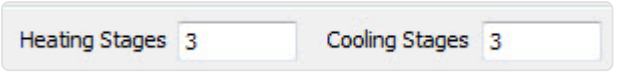
Off	Off	Off	Off	Off	Off	Off
Off	Off	Off	Off	Off	Off	Off
Off	Off	Off	Off	Off	Off	Off
Off	Off	Off	Off	Off	Off	Off
Off	Off	Off	Off	Off	Off	Off
Closed	Closed	Closed	Closed	Closed	Closed	Closed
Closed	Closed	Closed	Closed	Closed	Closed	Closed

Here are the states for the various outputs to each stage of cooling, heating and coasting. Since we have selected the OFF mode at tab18, all the outputs will be OFF with the system is set to OFF mode.

AUTO Mode

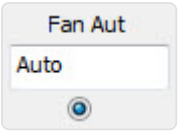


Heating/Cooling Stages



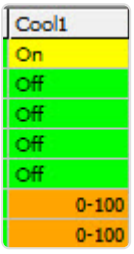
Here we set how many stages of heating and cooling the system will have. Since this is a three speed fan we can set three heating/cooling at this tab.If we set 2 or other number of heating and cooling stages,there will be corresponding quantity of tabs.

Fan Aut



Now select the Auto mode. The state of each of the outputs has been set to on, off or modulating on all of the various states.

Cool1



In the first stage of cooling, output1 is on for low speed fan operation and the cooling valve is modulating open from 0-50%.

Cool2

Cool2
On
Off
Off
Off
Off
Closed
0-100

In stage 2 of cooling, the medium fan speed is on and the valve is opening up from 50 to 100%.

Cool3

Cool3
Off
Off
On
On
Off
50-100
Closed

In stage 3 cooling the high speed fan is on and the valve is set once again to modulate from 50 to 100% .

Heating Stage

2	Heating	PID1	-	Off	On	Off	Off	Off	Off	Off

The same sequence is set up to the right of the table for the three stages of heating. The heating valve modulates open as shown and the cooling valve is closed for all stages of heating.

Variable Outputs State

Off	Off	On	Off	On	On	Off
Off	On	Off	Off	Off	Off	Off
On	Off	Off	Off	Off	Off	On
Off	Off	Off	Off	Off	Off	On
On	On	Off	Off	Off	Off	Off
Closed	Closed	Closed	Closed	0-100	Closed	50-100
50-100	50-100	Open	Closed	0-100	0-100	Closed

Each cell represents the state of a particular output at a certain stage of heating, cooling or coasting. In the coasting stage, all the outputs are off.

ON Mode

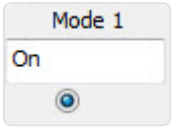


Heating/Cooling Stage



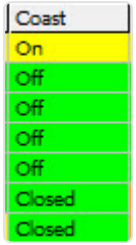
Here we set how many stages of heating and cooling the system will have. Since this is a three speed fan we can set three heating/cooling at this tab.If we set 2 or other number of heating and cooling stages,there will be corresponding quantity of tabs.

ON Mode



Now select the ON mode, this is generally used if you would like to allow the user to manually turn on the fan to get some fresh air in the zone.

Coast



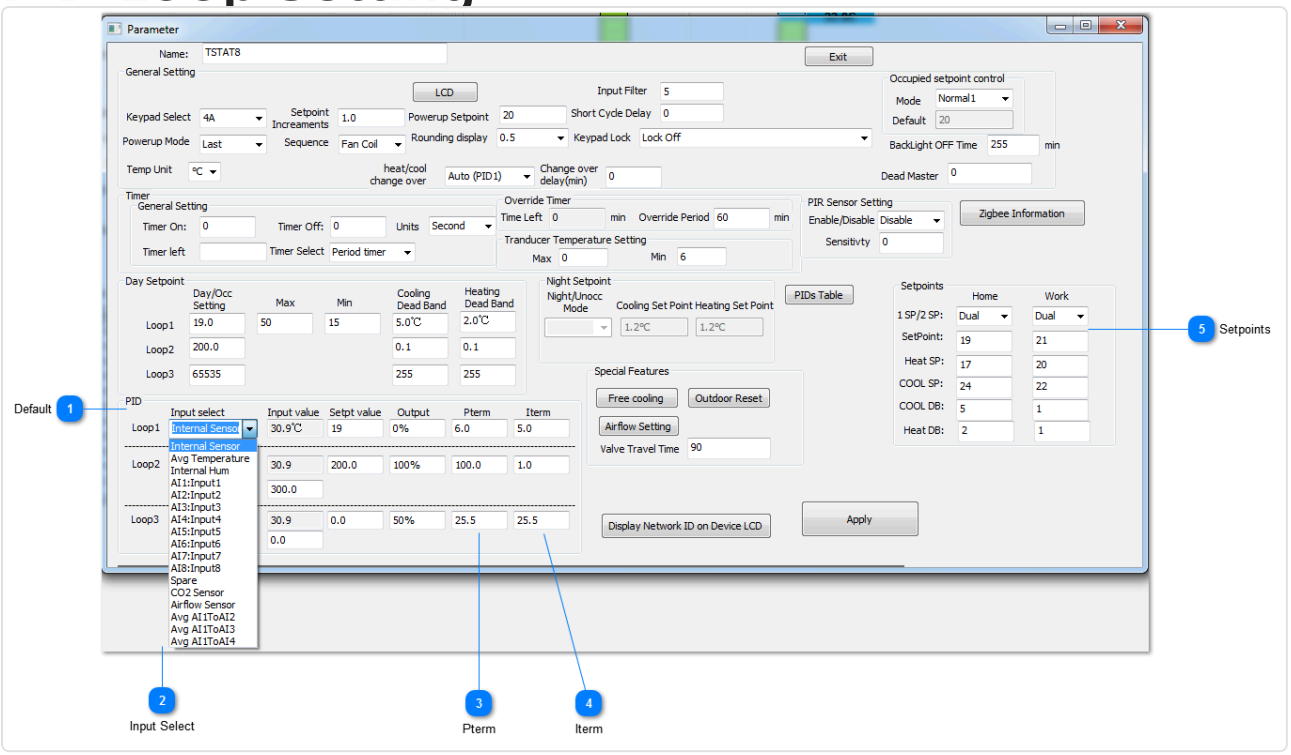
The stages of heating and cooling are set up exactly as everything was done in the Auto mode,the only difference is the Coast tab, you can see that the fan will be on in the coasting mode. This means even when the setpoint is satisfied at least the low speed fan will be on and the heating & cooling valves will be closed.

Variable Outputs State

On	On	On	On	On	On	On
Off	Off	Off	Off	Off	Off	Off
Off	Off	Off	Off	Off	Off	Off
Off	Off	Off	Off	Off	On	On
On	Off	Off	Off	Off	Off	Off
Closed	Closed	Closed	Closed	0-100	Closed	50-100
50-100	50-100	Open	Closed	0-100	0-100	Closed

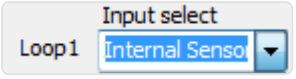
In all other stages of heating stage three on over to cooling stage3, the fan and valve are sequenced just like they were in the auto table.

PID Loop Setting



PID loop

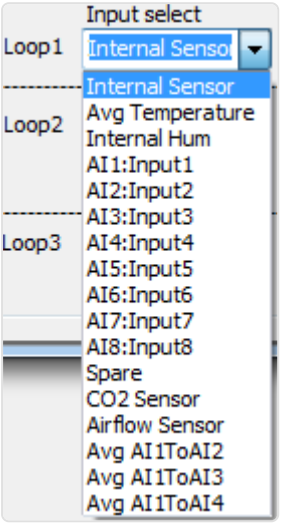
Default



Now we set up the PID loop, for the most part you can leave the settings at their default.

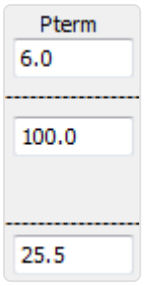
You can select which of the inputs will provide feedback for the PID loop, the default is the internal temperature sensor.

Input Select



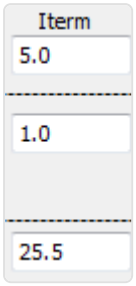
For the PID feedback you can select from many options such as the average of some sensors, any of the 8 external sensors or the internal humidity or C02 sensor if it has one.

Pterm



This is the PID proportional setting. The default value of 6 means that the PID loop will output a full response when the temperature error is 6 degrees, in this case celcius, away from setpoint. So if the setpoint were 20C, the maximum heating to the maximum cooling will occur over a span of 6 degrees. When we□re three degrees below setpoint we□ll be in stage 3 heating and when we□re 3 degrees above setpoint the PID will be at stage 3 cooling.

Iterm



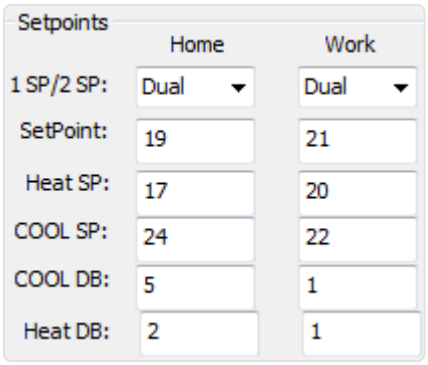
This is the integral term. This is a nudge factor so that if the temperature is hovering a little ways below the setpoint for a long period eventually the integral term will accumulate some error over time and begin to bring on the next stage of heating. During commissioning this term can be set to zero because it can wind-up and cause confusion. For example the room is just below the setpoint by half a degree or so you would think the unit will be in stage1 heat yet it is running at high speed. It is the integral term which has wound up and is calling for stage three. When commissioning is done you can enter something for the I term to get better PID loop action, the default value of 5 is reasonable and means you can get an extra 5% of PID action for every degC □ hour of accumulated error.

A small value for the P term means the PID loop will be more sensitive to a deviation from setpoint. A small term for the I term means the PID loop will be lazy over time.

Small P = hyperactive

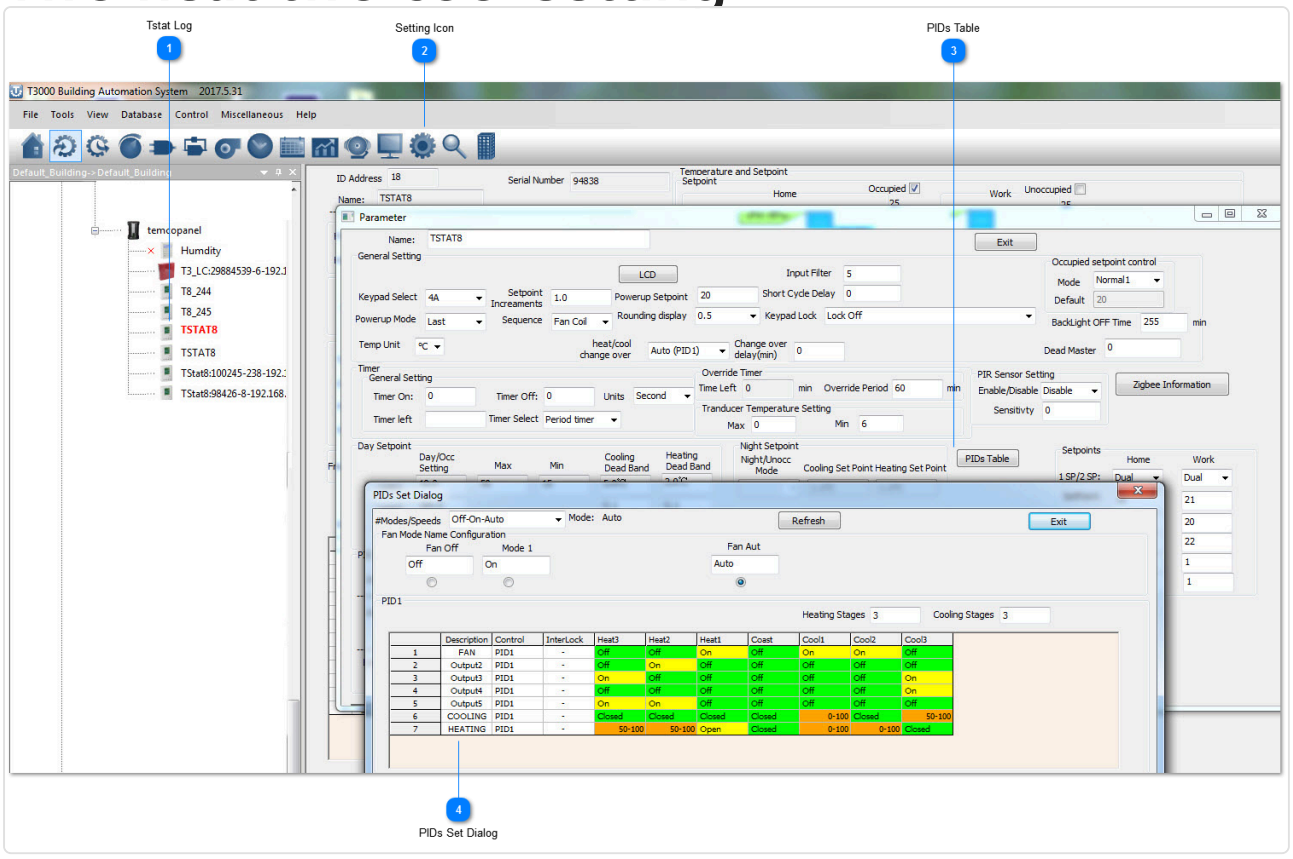
Small I = lazy, over time.

Setpoints



Finally set up the setpoints. Dual setpoints operate like a residential thermostat where you can set up a separate heating and cooling setpoint. Single setpoints are more for commercial settings where you would like to keep things simple and let the user adjust only a single setpoint up or down. The HOME and WORK columns show the heating and cooling setpoints for occupied (home) and unoccupied modes (work). There are other modes which we need to add to this dialog, SLEEP and AWAY which are common with residential thermostats. In single setpoint mode there is only one setpoint we talk about, so the heating setpoint is calculated from the "setpoint" minus the heating deadband. Similarly, the cooling setpoint is equal to the "setpoint" plus the cooling deadband.

Two heat two cool setting



Tstat Log



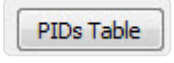
Click on the gear icon to get to the advanced settings.

Setting Icon



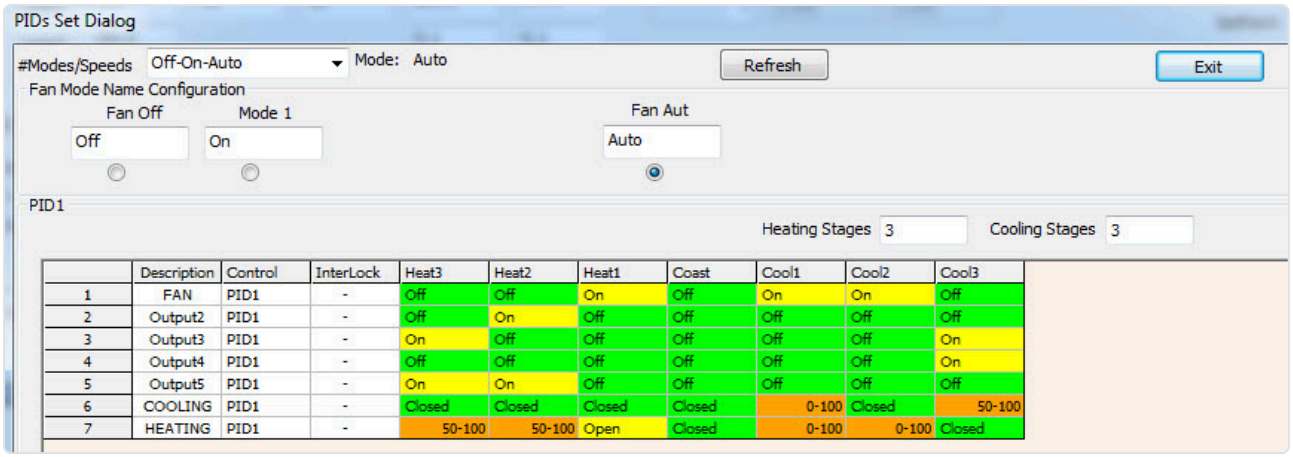
Click to get to the PID tables, this is where we assign outputs to act over each stage of heating and cooling.

PIDs Table



: Insert description text here...

PIDs Set Dialog

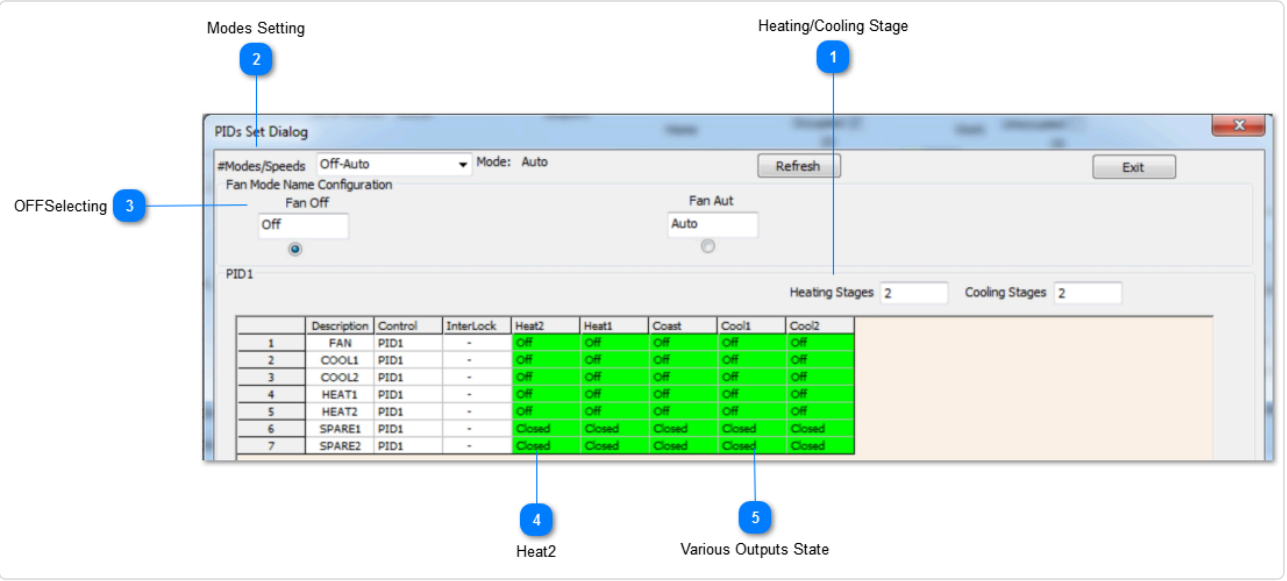


	Description	Control	InterLock	Heat3	Heat2	Heat1	Coast	Cool1	Cool2	Cool3
1	FAN	PID1	-	Off	Off	On	Off	On	On	Off
2	Output2	PID1	-	Off	On	Off	Off	Off	Off	Off
3	Output3	PID1	-	On	Off	Off	Off	Off	Off	On
4	Output4	PID1	-	Off	Off	Off	Off	Off	Off	On
5	Output5	PID1	-	On	On	Off	Off	Off	Off	Off
6	COOLING	PID1	-	Closed	Closed	Closed	Closed	0-100	Closed	50-100
7	HEATING	PID1	-	50-100	50-100	Open	Closed	0-100	0-100	Closed

PIDs Set Dialog shows the details of the setting.

The contents of Two heat two cool setting

OFF Mode



Heating/Cooling Stage



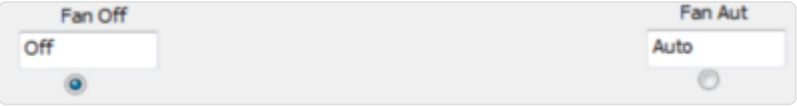
Set the number of stages of heating and cooling.

Modes Setting



You can set the number of modes at the tab, two modes have been selected here with the default names as Off and Auto.

OFFSelecting



You can rename them by editing the names at the tab.Next we□ll edit the Off table by selecting the radio button. The grid shown represents the outputs when the stat is in the off mode.

Heat2

Heat2
Off
Off
Off
Off
Off
Closed
Closed

The center column represents the coasting mode, everything is off there as well.

Various Outputs State

Heat2	Heat1	Coast	Cool1	Cool2
Off	Off	Off	Off	Off
Off	Off	Off	Off	Off
Off	Off	Off	Off	Off
Off	Off	Off	Off	Off
Off	Off	Off	Off	Off
Closed	Closed	Closed	Closed	Closed
Closed	Closed	Closed	Closed	Closed

Each row is one output and the columns represent the outputs in each of the stages of heating and cooling. In this example, everything is off for all 2 stages of heat and two cool.

AUTO Mode

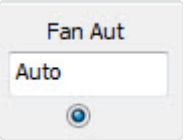


Heating/Cooling Stage



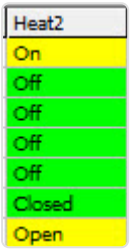
Set the number of stages of heating and cooling.

Fan Aut



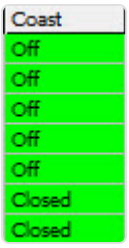
Fill in the grid for the Auto mode.

Heat2



Heat2 will be on in the column called Heat2 and off for the other stages and so on. The fan is on all the time as shown by the FAN row with all yellow entries.

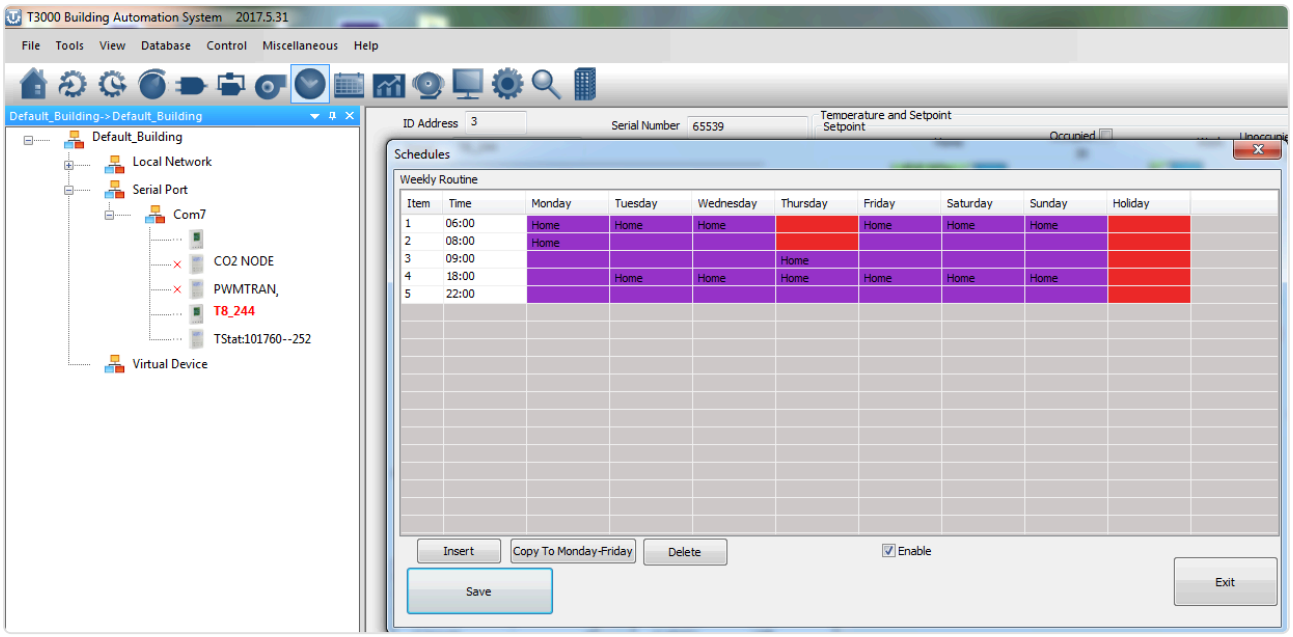
Coast

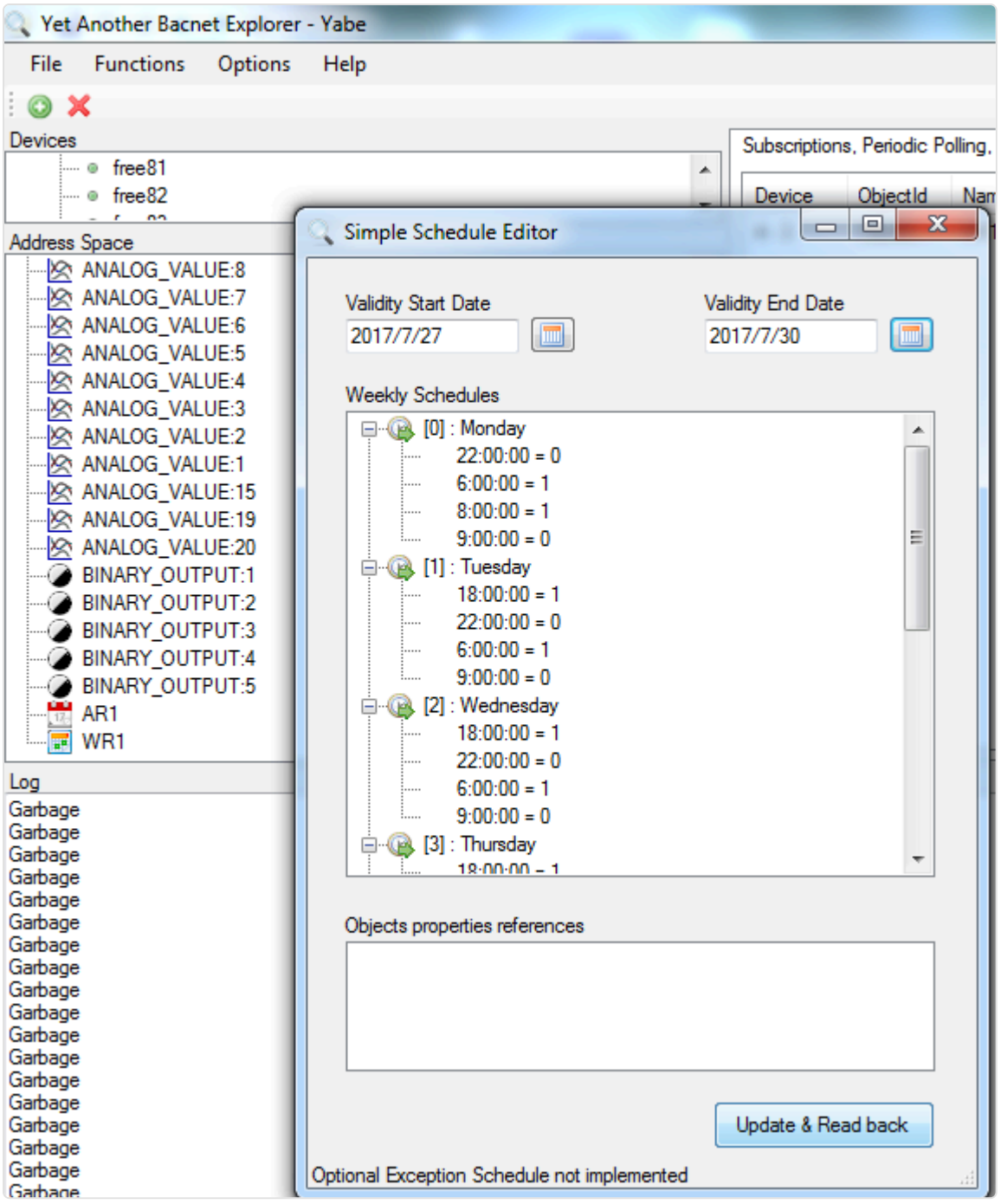


You could optionally turn the fan off during coasting by setting the cell to OFF.

Schedules

The schedules of Tstat8 can be managed using T3000 software and Bacnet.Select Modbus protocol when you use T3000,and Bacnet protocol when you use Bacnet.


T3000 Building Automation System 2017.5.31
File Tools View Database Control Miscellaneous Help
Default_Building-> Default_Building
Default_Building
Local Network
Serial Port
Com7
CO2 NODE
PWMTRAN,
T8_244
TStat:101760--252
Virtual Device
ID Address 3
Serial Number 65539
Temperature and Setpoint
Setpoint
Occupied
Schedules
Weekly Routine
Item Time Monday Tuesday Wednesday Thursday Friday Saturday Sunday Holiday
1 06:00 Home Home Home Home Home Home
2 08:00 Home
3 09:00 Home
4 18:00 Home Home Home Home Home Home
5 22:00
Insert
Copy To Monday-Friday
Delete
Enable
Save
Exit



The contents of Schedules

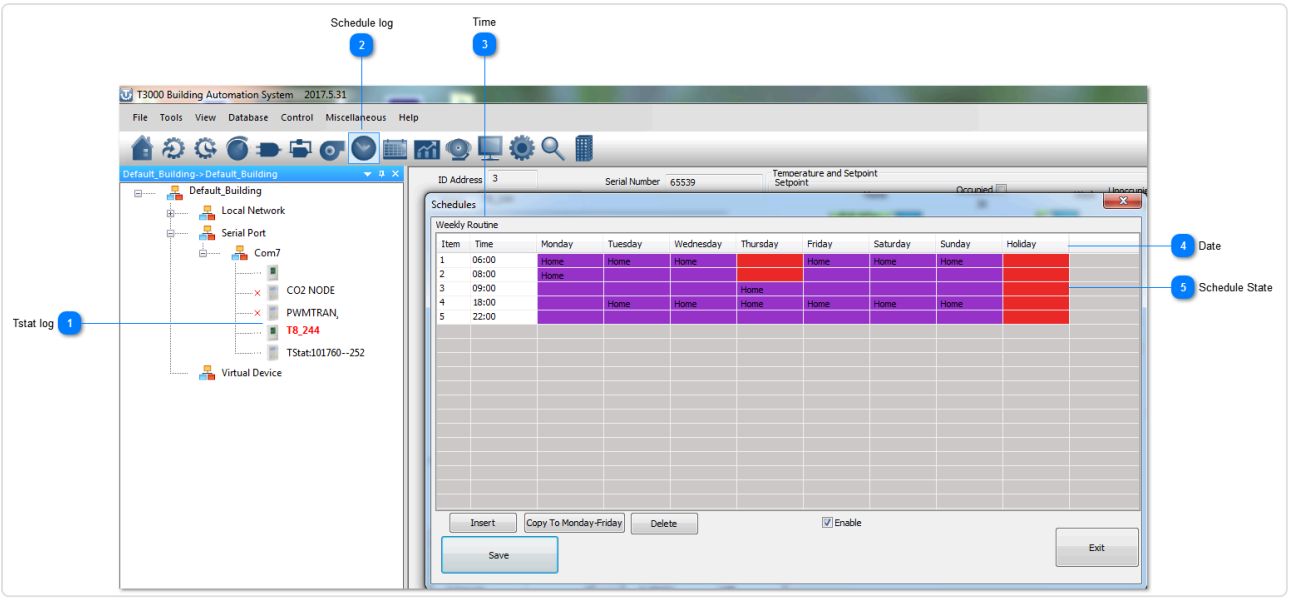
Managing schedules in T3000

The schedules can be managed using T3000 software.Below are the steps:

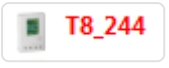
Step1.Visit 107.170.34.189/ftp/software/9TstatSoftware.zip ,download T3000 software and install it;

Step2.Plug Tstat8 in power,connect Tstat8 to PC via RS485 or Ethernet;

Step3.Start T3000 software,click  to scan,then you can find Tstat8 as below.



Tstat log



Click to select the thermostat.

Schedule log



Click to show schedule details.

Time

Weekly Routine

Item	Time
1	06:00
2	08:00
3	09:00
4	18:00
5	22:00

This is the time list.

Date

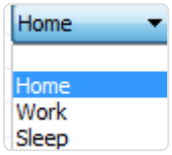
Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	Holiday

The date diaplays from Monday to Sunday and Holiday.

Schedule State

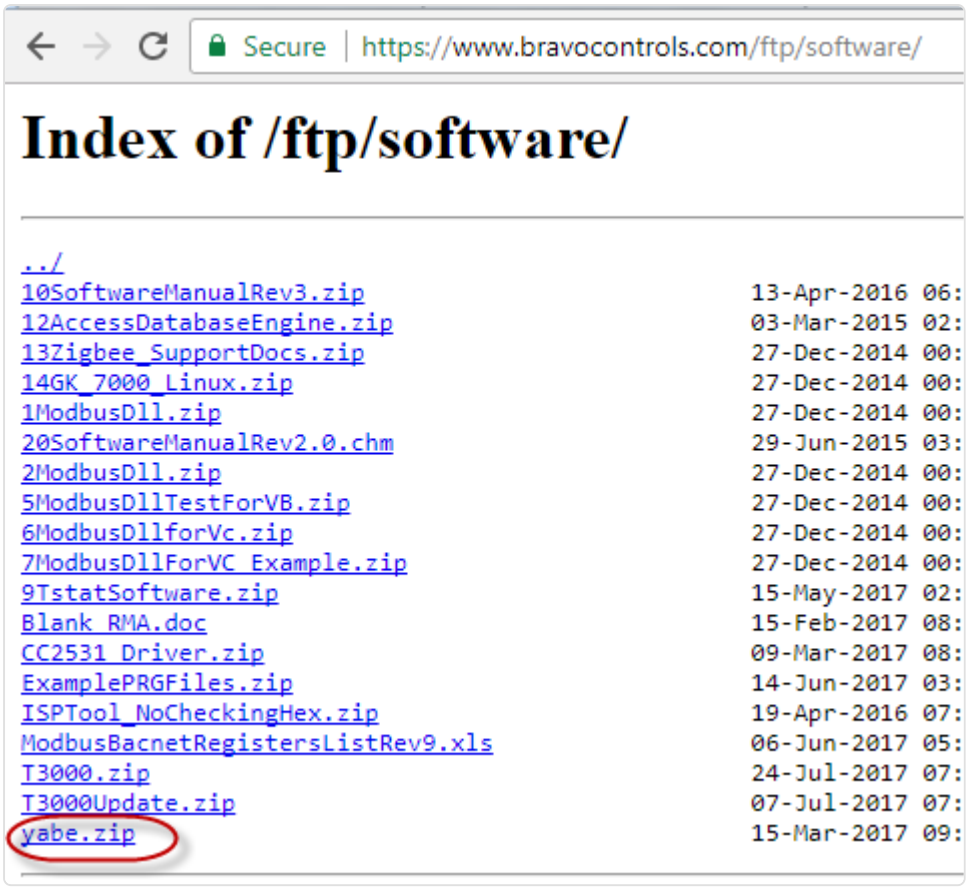
Home	Home	Home		Home	Home	Home	
Home							
			Home				
	Home	Home	Home	Home	Home	Home	

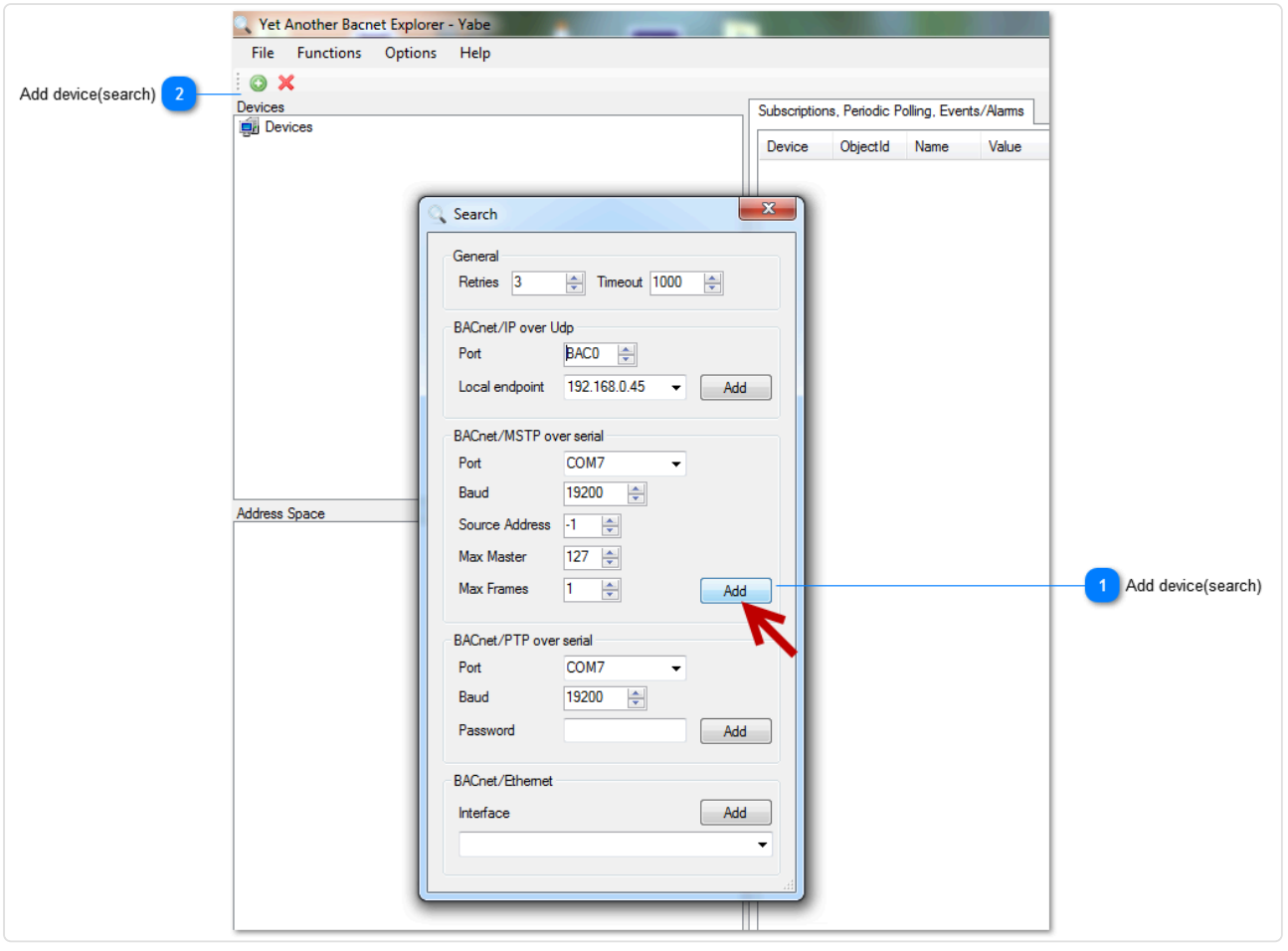
The tab shows the schedule state:Home/Work/Sleep.



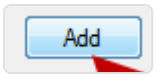
Managing schedules using Bacnet

The schedules can be managed using Bacnet.Download Yabe software as the link:107.170.34.189/ftp/software/yabe.zip and install it.Connect Tstat8 to the computer,select Bacnet protocol.Start the Yabe software,add the device.


Secure | https://www.bravocontrols.com/ftp/software/
Index of /ftp/software/
../
10SoftwareManualRev3.zip 13-Apr-2016 06:
12AccessDatabaseEngine.zip 03-Mar-2015 02:
13Zigbee_SupportDocs.zip 27-Dec-2014 00:
14GK_7000_Linux.zip 27-Dec-2014 00:
1ModbusDll.zip 27-Dec-2014 00:
20SoftwareManualRev2.0.chm 29-Jun-2015 03:
2ModbusDll.zip 27-Dec-2014 00:
5ModbusDllTestForVB.zip 27-Dec-2014 00:
6ModbusDllforVc.zip 27-Dec-2014 00:
7ModbusDllForVC_Example.zip 27-Dec-2014 00:
9TstatSoftware.zip 15-May-2017 02:
Blank_RMA.doc 15-Feb-2017 08:
CC2531_Driver.zip 09-Mar-2017 08:
ExamplePRGFiles.zip 14-Jun-2017 03:
ISPTool_NoCheckingHex.zip 19-Apr-2016 07:
ModbusBacnetRegistersListRev9.xls 06-Jun-2017 05:
T3000.zip 24-Jul-2017 07:
T3000Update.zip 07-Jul-2017 07:
yabe.zip 15-Mar-2017 09:


Add device(search)
2
Yet Another Bacnet Explorer - Yabe
File
Functions
Options
Help
Devices
Devices
Subscriptions, Periodic Polling, Events/Alarms
Device
ObjectId
Name
Value
Address Space
Search
General
Retries
3
Timeout
1000
BACnet/IP over Udp
Port
BAC0
Local endpoint
192.168.0.45
Add
BACnet/MSTP over serial
Port
COM7
Baud
19200
Source Address
-1
Max Master
127
Max Frames
1
Add
BACnet/PTP over serial
Port
COM7
Baud
19200
Password
Add
BACnet/Ethernet
Interface
Add
1
Add device(search)

Add device(search)



Click to add via BACnet/MSTP .

Add device(search)

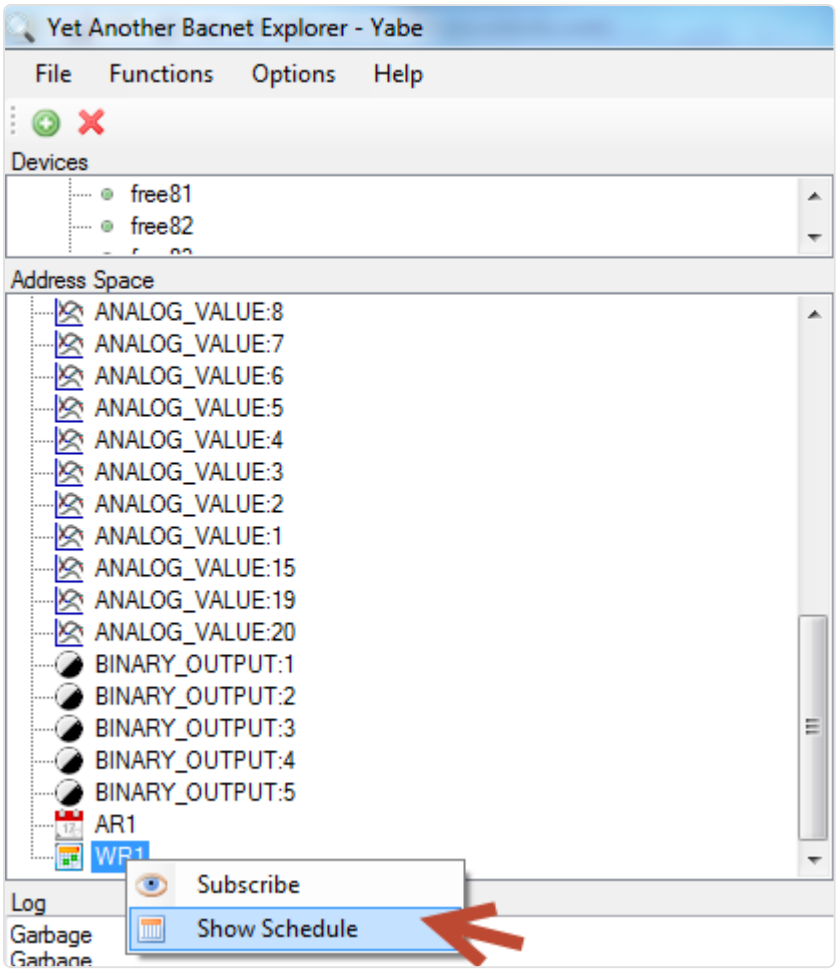


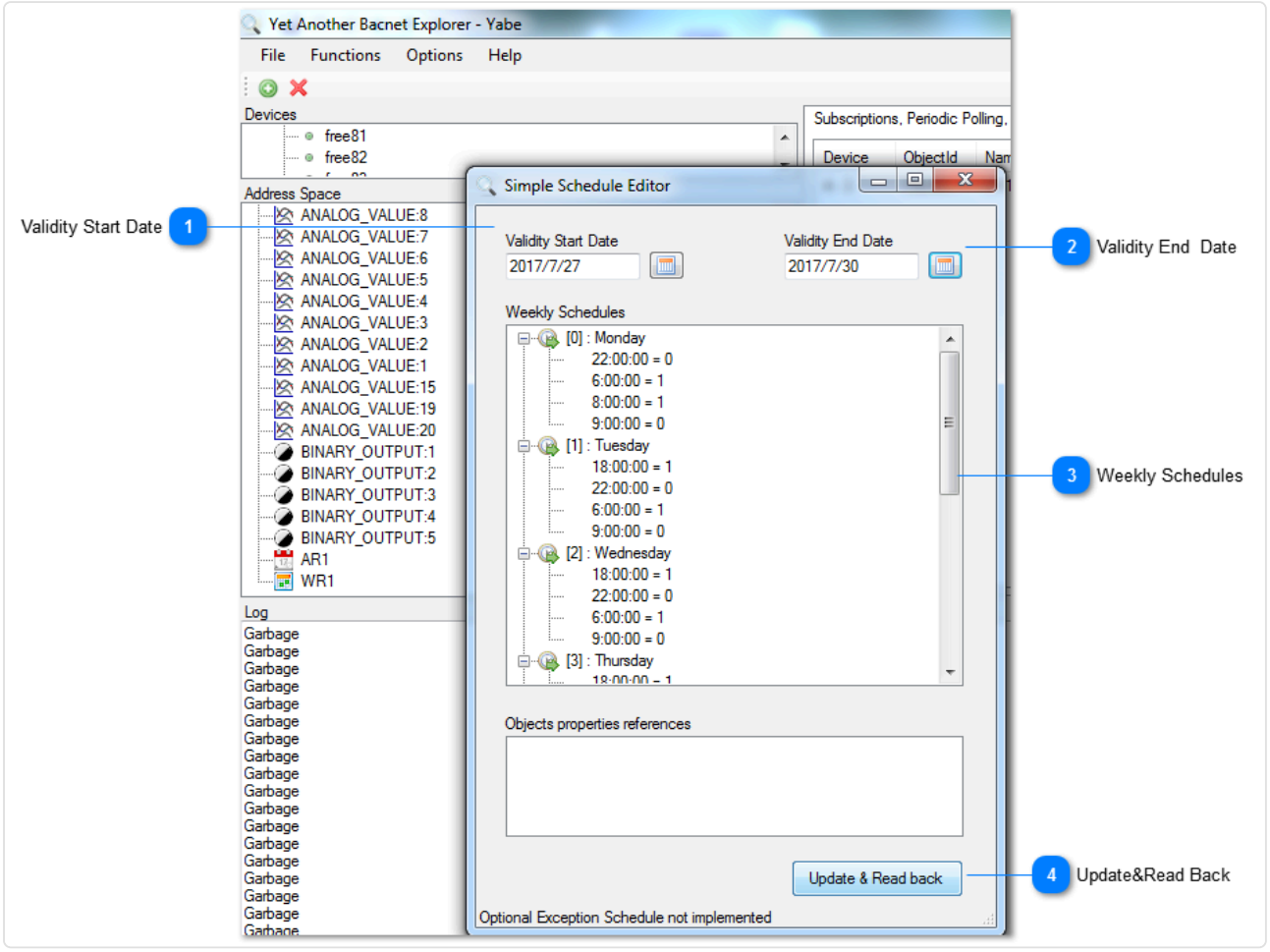
Click to add device.

The contents of Managing schedules using Bacnet

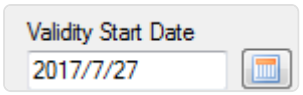
Weekly routines schedule

Right click □WR1□ log (weekly routines) to set up the weekly schedule date.For the weekly routine, if status = 1, means unit will go to occupied and if status = 0, means unit will go to unoccupied. When the time is set up, click □Update & Read back□ button to save and read the setting back.


Yet Another Bacnet Explorer - Yabe
File
Functions
Options
Help
Devices
free81
free82
Address Space
ANALOG_VALUE:8
ANALOG_VALUE:7
ANALOG_VALUE:6
ANALOG_VALUE:5
ANALOG_VALUE:4
ANALOG_VALUE:3
ANALOG_VALUE:2
ANALOG_VALUE:1
ANALOG_VALUE:15
ANALOG_VALUE:19
ANALOG_VALUE:20
BINARY_OUTPUT:1
BINARY_OUTPUT:2
BINARY_OUTPUT:3
BINARY_OUTPUT:4
BINARY_OUTPUT:5
AR1
WR1
Subscribe
Show Schedule
Log
Garbage

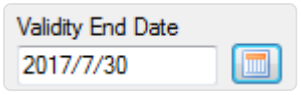

Yet Another Bacnet Explorer - Yabe
File
Functions
Options
Help
Devices
free81
free82
Address Space
ANALOG_VALUE:8
ANALOG_VALUE:7
ANALOG_VALUE:6
ANALOG_VALUE:5
ANALOG_VALUE:4
ANALOG_VALUE:3
ANALOG_VALUE:2
ANALOG_VALUE:1
ANALOG_VALUE:15
ANALOG_VALUE:19
ANALOG_VALUE:20
BINARY_OUTPUT:1
BINARY_OUTPUT:2
BINARY_OUTPUT:3
BINARY_OUTPUT:4
BINARY_OUTPUT:5
AR1
WR1
Log
Garbage
Subscriptions, Periodic Polling,
Device
ObjectId
Simple Schedule Editor
Validity Start Date
2017/7/27
Validity End Date
2017/7/30
Weekly Schedules
[0] : Monday
22:00:00 = 0
6:00:00 = 1
8:00:00 = 1
9:00:00 = 0
[1] : Tuesday
18:00:00 = 1
22:00:00 = 0
6:00:00 = 1
9:00:00 = 0
[2] : Wednesday
18:00:00 = 1
22:00:00 = 0
6:00:00 = 1
9:00:00 = 0
[3] : Thursday
Objects properties references
Update & Read back
Optional Exception Schedule not implemented
Validity Start Date
1
2
Validity End Date
3
Weekly Schedules
4
Update&Read Back

Validity Start Date



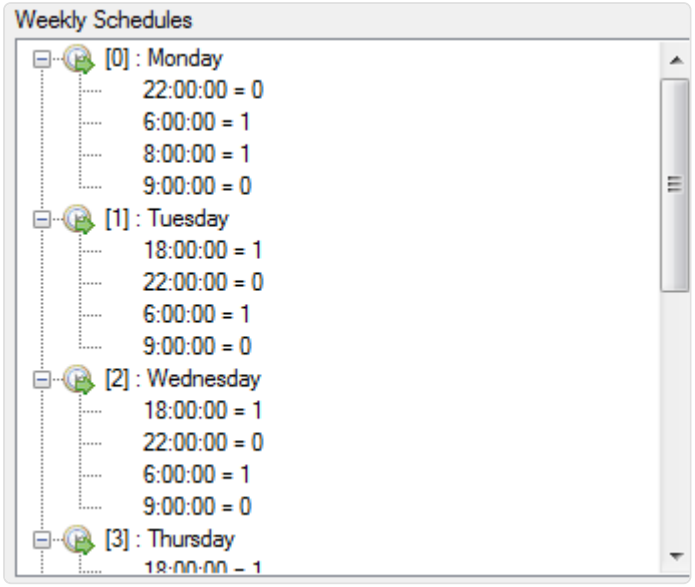
Click to set the start date.

Validity End Date



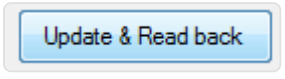
Click to set the End date.

Weekly Schedules



This tab shows the Weekly Schedules details.

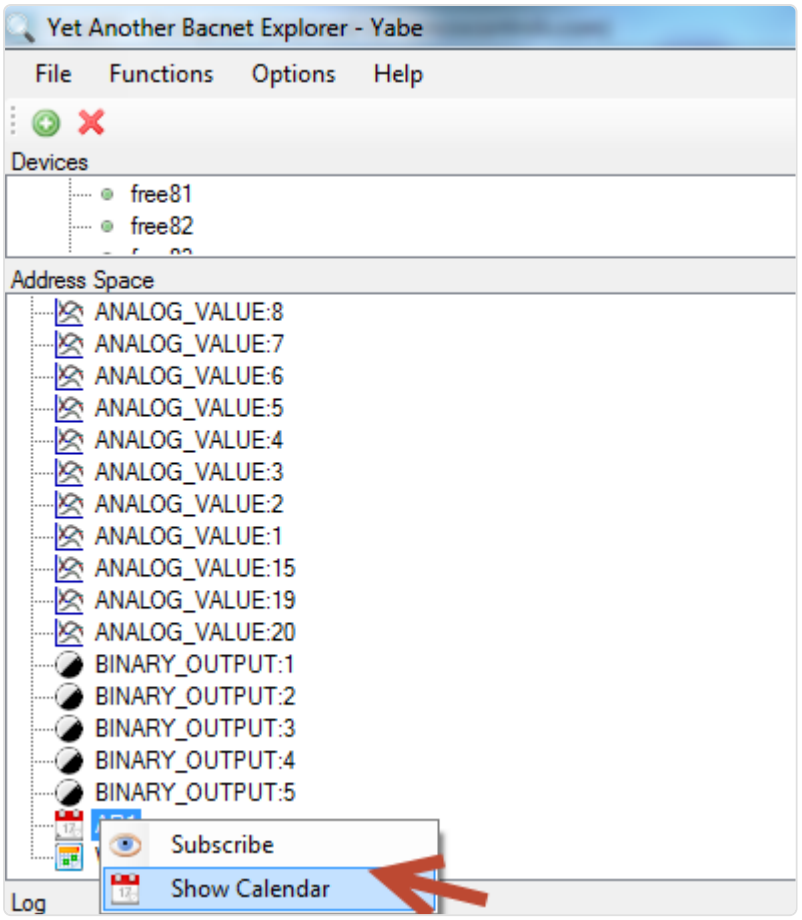
Update&Read Back

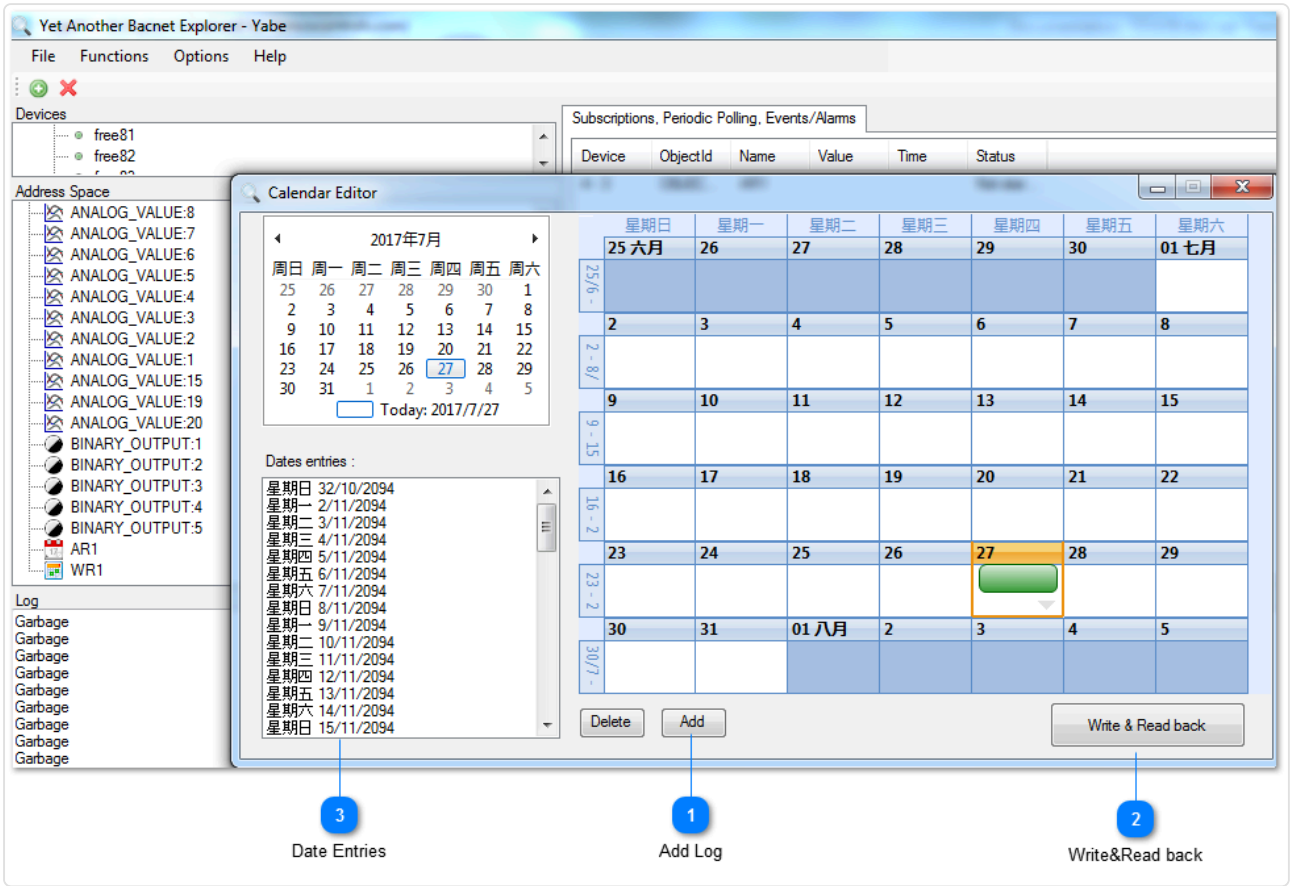


Click the log to update &Read Back.

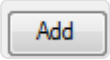
Annual routine schedule

Step4.Right click □AR1□ log to set up annual routine date.You can set up the annual date from this tab.


Yet Another Bacnet Explorer - Yabe
File
Functions
Options
Help
Devices
free81
free82
Address Space
ANALOG_VALUE:8
ANALOG_VALUE:7
ANALOG_VALUE:6
ANALOG_VALUE:5
ANALOG_VALUE:4
ANALOG_VALUE:3
ANALOG_VALUE:2
ANALOG_VALUE:1
ANALOG_VALUE:15
ANALOG_VALUE:19
ANALOG_VALUE:20
BINARY_OUTPUT:1
BINARY_OUTPUT:2
BINARY_OUTPUT:3
BINARY_OUTPUT:4
BINARY_OUTPUT:5
Subscribe
Show Calendar
Log

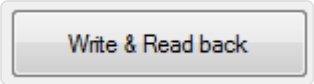

Yet Another Bacnet Explorer - Yabe
File
Functions
Options
Help
Devices
free81
free82
Address Space
ANALOG_VALUE:8
ANALOG_VALUE:7
ANALOG_VALUE:6
ANALOG_VALUE:5
ANALOG_VALUE:4
ANALOG_VALUE:3
ANALOG_VALUE:2
ANALOG_VALUE:1
ANALOG_VALUE:15
ANALOG_VALUE:19
ANALOG_VALUE:20
BINARY_OUTPUT:1
BINARY_OUTPUT:2
BINARY_OUTPUT:3
BINARY_OUTPUT:4
BINARY_OUTPUT:5
AR1
WR1
Log
Garbage
Subscriptions, Periodic Polling, Events/Alarms
Device
ObjectId
Name
Value
Time
Status
Calendar Editor
2017年7月
Today: 2017/7/27
Dates entries :
星期日 32/10/2094
星期一 2/11/2094
星期二 3/11/2094
星期三 4/11/2094
星期四 5/11/2094
星期五 6/11/2094
星期六 7/11/2094
星期日 8/11/2094
星期一 9/11/2094
星期二 10/11/2094
星期三 11/11/2094
星期四 12/11/2094
星期五 13/11/2094
星期六 14/11/2094
星期日 15/11/2094
Delete
Add
Write & Read back
3
Date Entries
1
Add Log
2
Write&Read back

Add Log



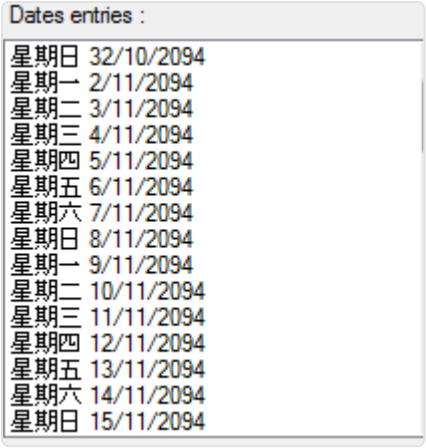
Click to add the date.

Write&Read back



Click to write & read back.

Date Entries



Click to choose the date.

Tstat8 Hardware

Highlights:

 8 universal inputs for external temperature, voltage, contacts, etc.

 5 relay outputs, rated at 24vac @ 2 amps each;2 analog outputs, 0-10V @ 100ma each.

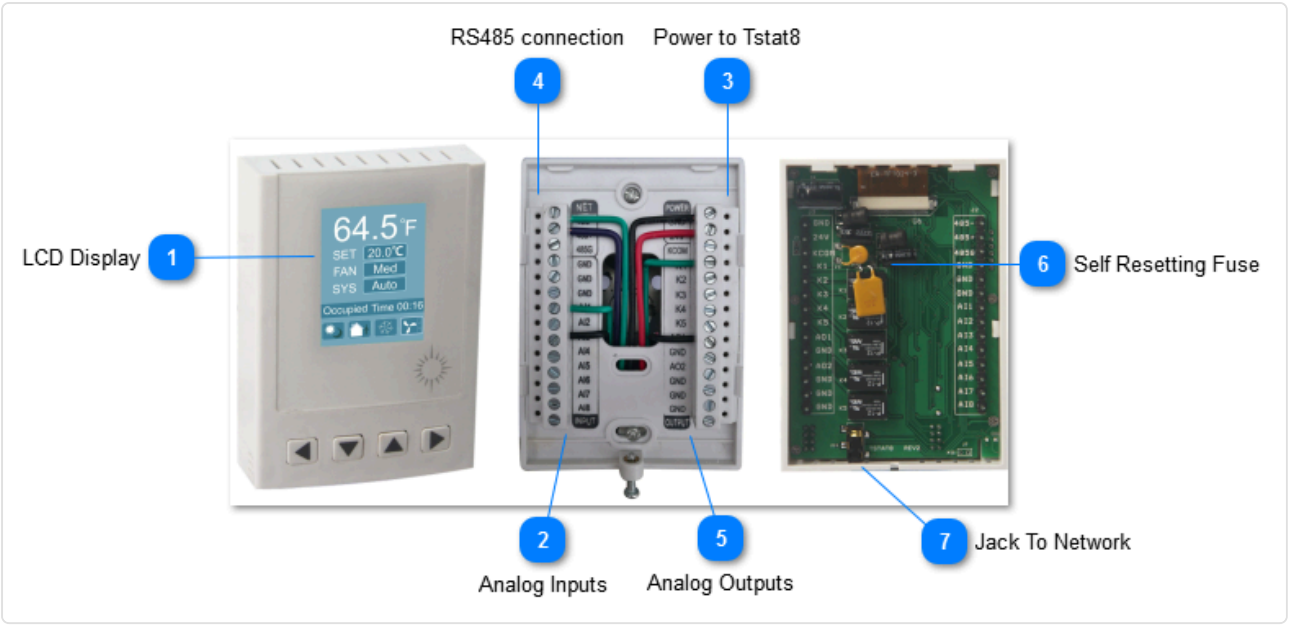
 Color LCD display,2.4 inch diagonal.

 On board clock with infiite life supercap battery backup.

 Uses a 32 bit Arm CPU with 12 bit analog resolution.

The contents of Tstat8 Hardware

Overview



LCD Display



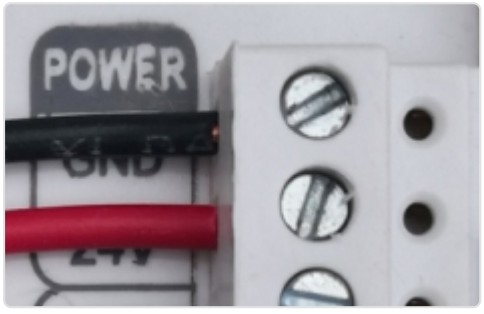
Color LCD display,2.4 inch diagonal

Analog Inputs



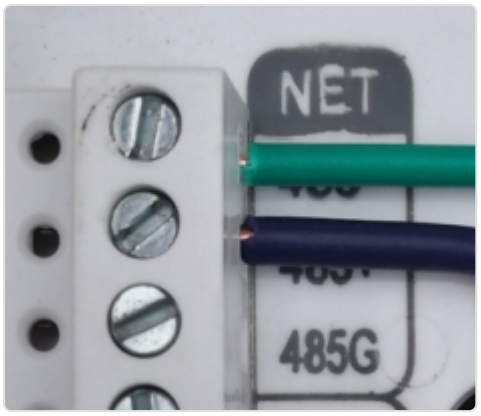
Analog inputs,8 universal inputs ,10k therm, contacts, 4-20ma, 0-5V,0-10V

Power to Tstat8



The power supply can be any 12V to 24V supply, AC or DC with capacity of approximately 2 watts. Surge and reverse polarity protected with slow blow self resetting fuse on board.

RS485 connection



RS485 connection,BACnet MS/TP and Modbus RTU protocols over RS485.

Analog Outputs

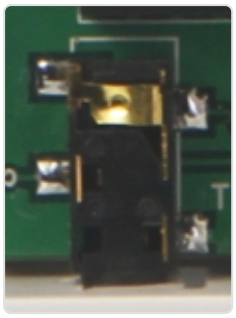


Analog outputs,5 relay outputs 24vac @ 2 amps;2 analog outputs 10V @100mA.

Self Resetting Fuse



Jack To Network



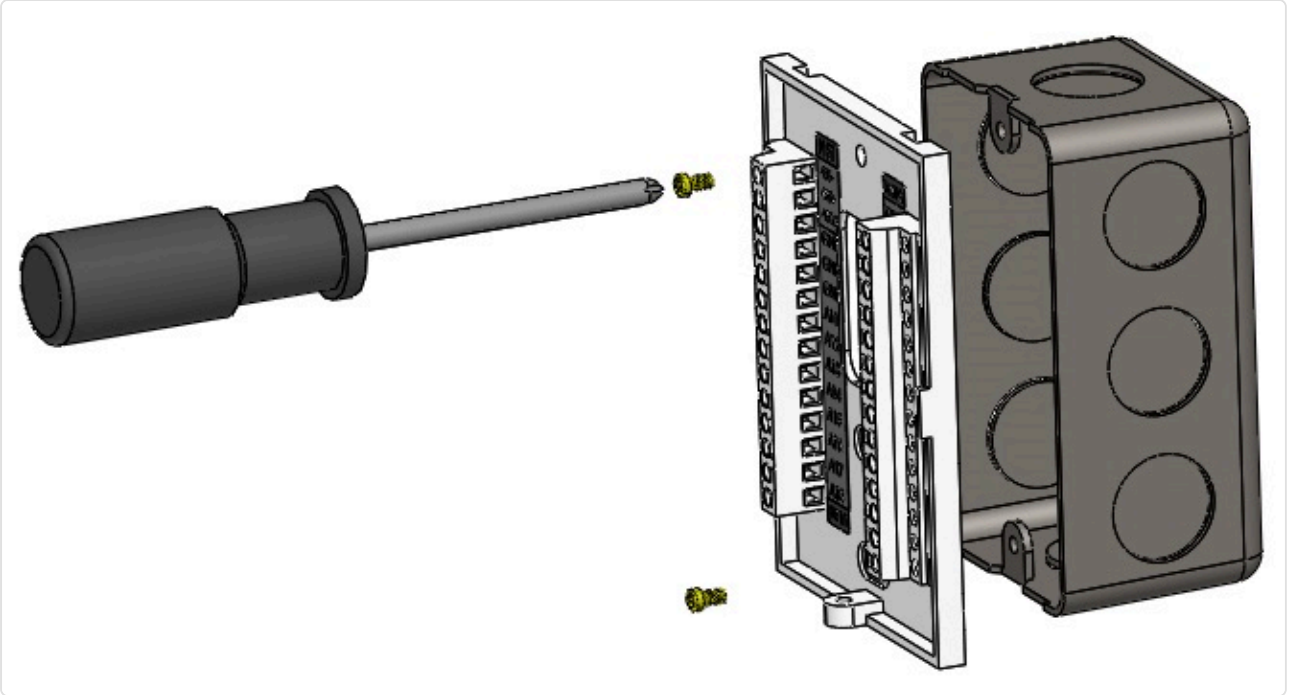
Installation

The Installation Mounting includes Wall Mount and/or Electrical Box Mount.You can install Tstat8 as following steps:

1)Unfasten the screw located at the base and lift off the front panel of the enclosure.



2)Fasten the screws on the backplate to the wall or electrical box.



3)Connect the wires to the terminals and re-attach the front panel.



**

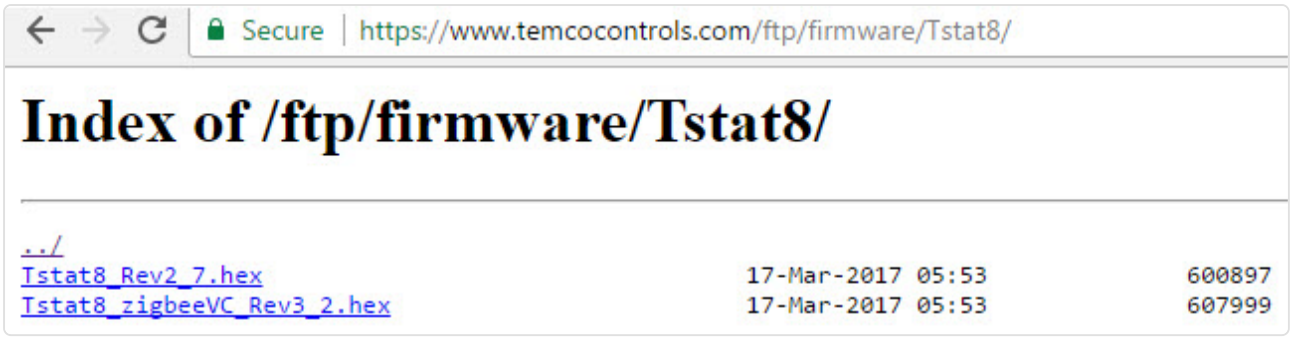
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FIRMWARE UPDATE, SINGLE DEVICE

Using the Tstat8 as an example

1> Download the latest firmware from temcocontrols.com ,or paste "http://www.temcocontrols.com/ftp/firmware/Tstat8" to your browser .

Select the version that you want to update to download .

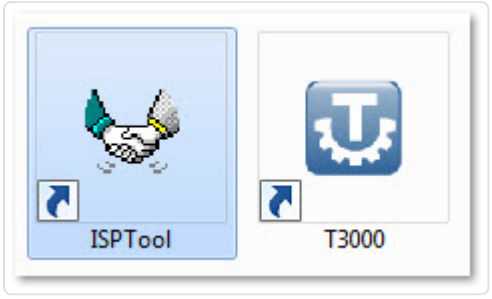


If you get the latest firmware from temco's engineer, don't need the step 1.

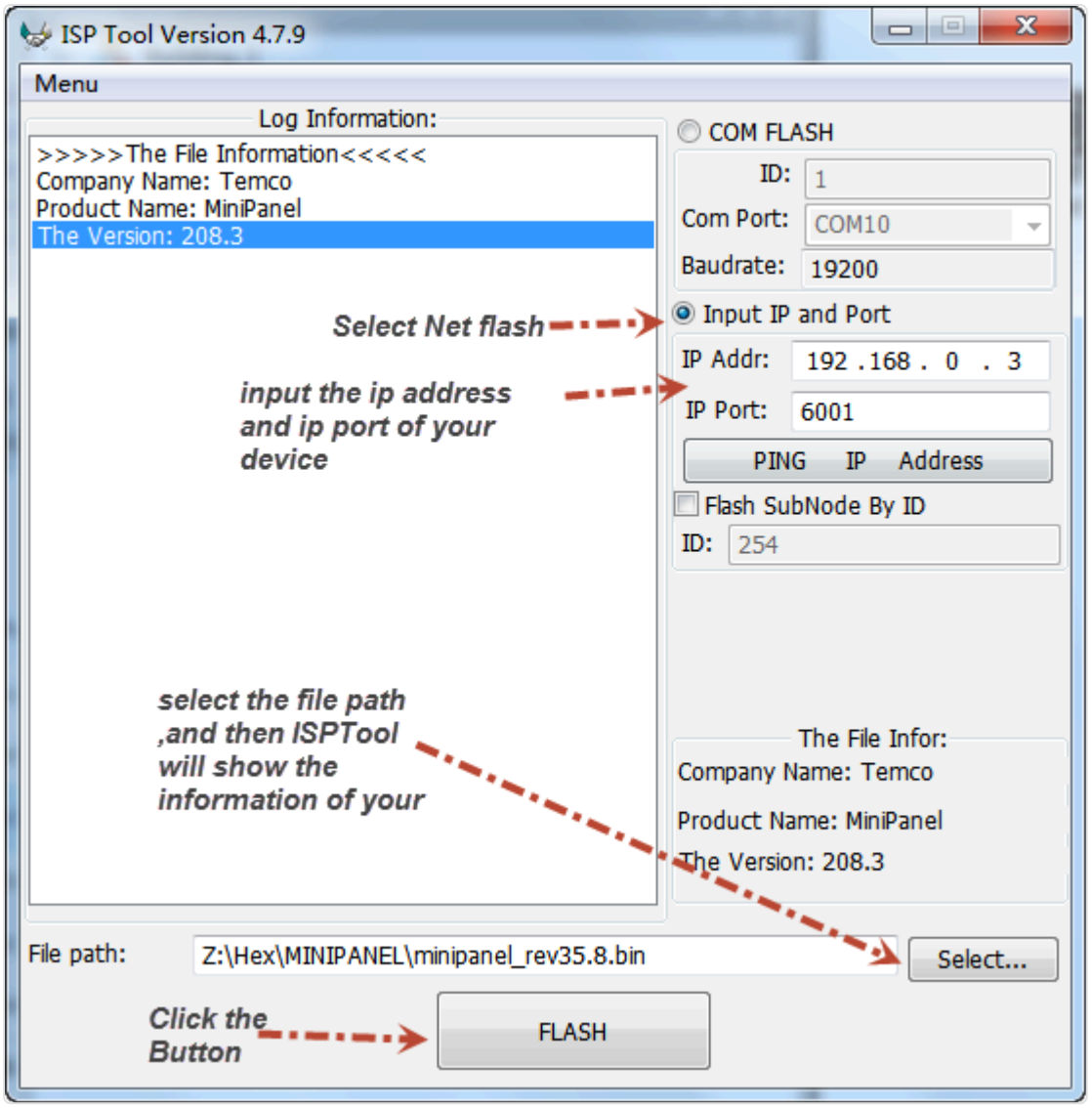
2> If your computer didn't install the T3000 ,please download T3000.zip .

If you have installed the T3000,don't need the step2.

3> Click the icon of ISPTool



and then show a window.



If you want to update other device ,using Com port ,please select "COM FLASH"

And then input the Modbus ID of your device .If you don't know your Modbus ID ,you can try flash the device ,using the 255.

select your Com Port .

and then just click the button of "FLASH"

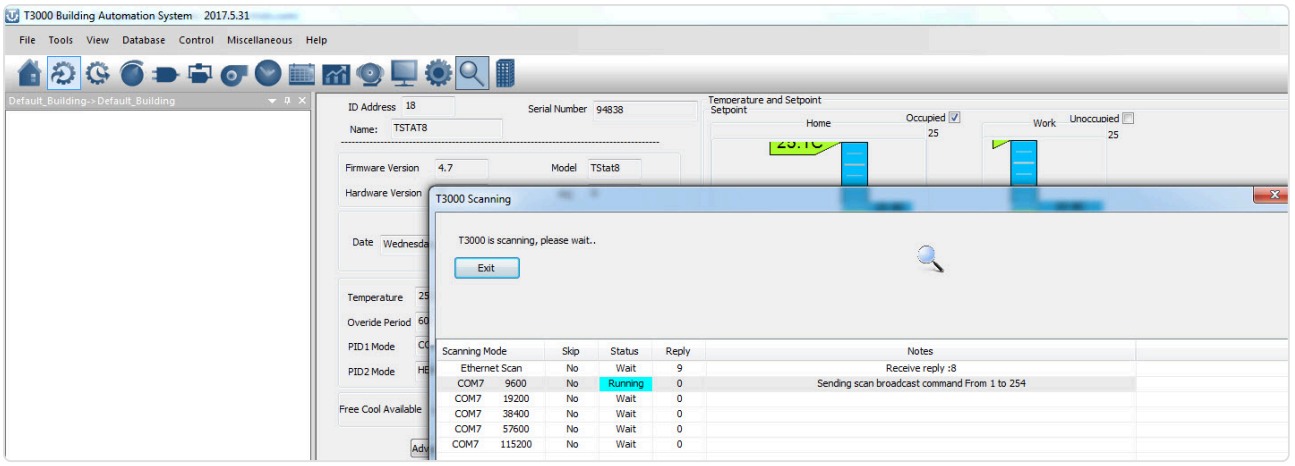
notes: If you don't know the ip address and ip port of the device, don't care about the two parameters .

ISPTool will show the right message in the blank area ,including the ip and ip port.

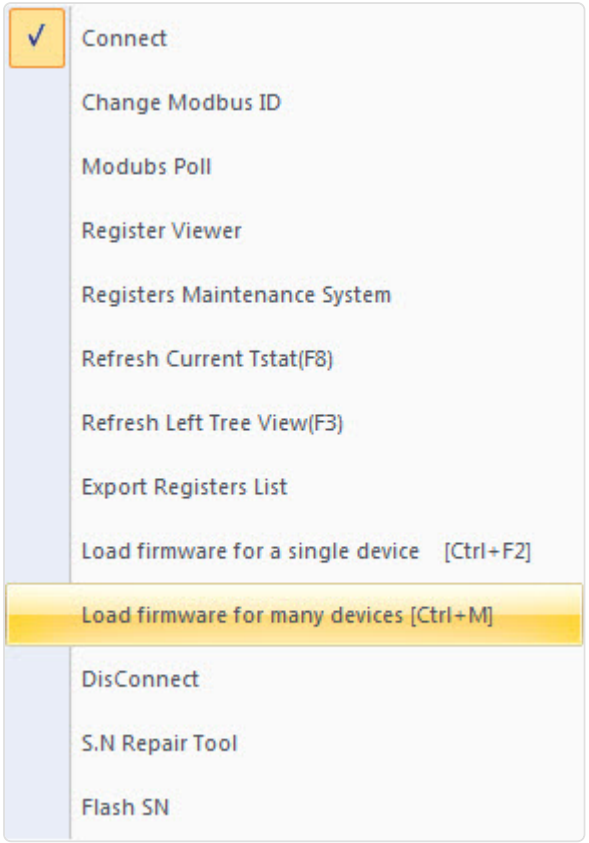
The contents of FIRMWARE UPDATE, SINGLE DEVICE

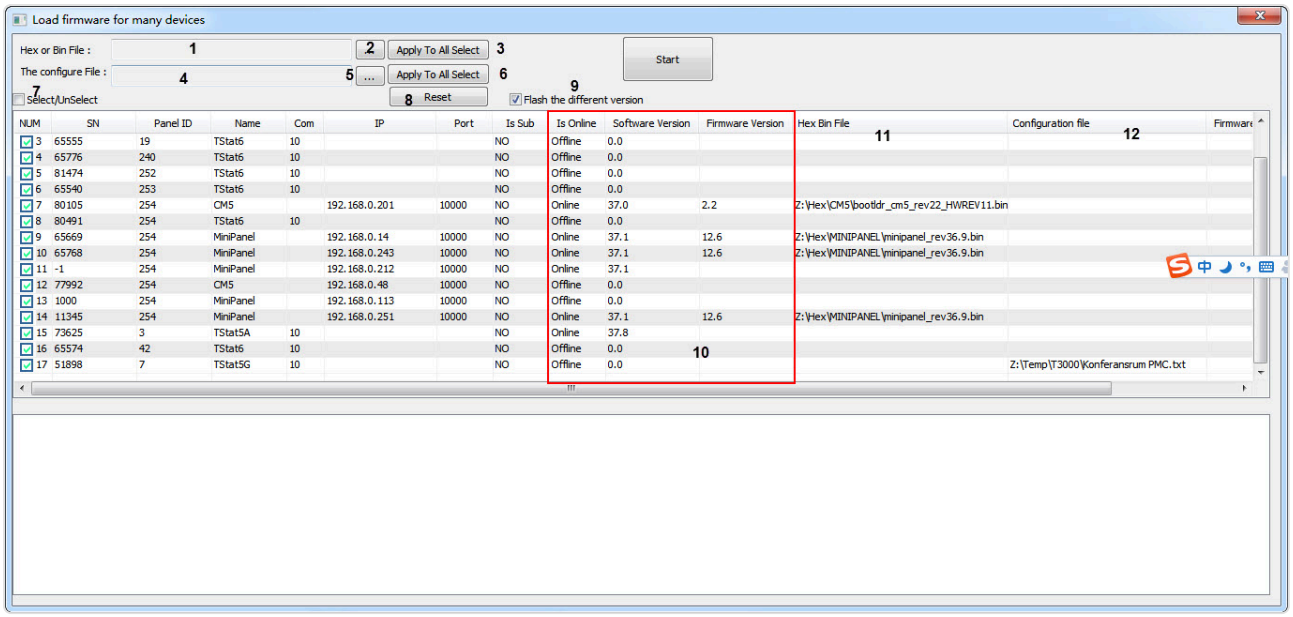
FIRMWARE UPDATES: BATCH LOADING

1.You must find your devices by T3000 , using "Scan"



2 Load firmware for many devices





1

show the path of a hex or bin file

2

click to select the path of a hex or bin file

3

"Apply To All Select" ,the path of the hex or bin(1) will automatically fill in the column of "Hex Bin File" , matching with the column of "Name"

4

show the path of a configuration file

5

click to select the path of a configuration file

6

"Apply To All Select" ,the path of the configuration file (4) will automatically fill in the column of "Configuration File" , matching with the column of "Name"

7

Select/Unselect

8

Reset ,to clear the column of "Hex Bin File" and the column of "Configuration File"

9

if selected ,T3000 will compare the "Software Version" with "Firmware Version", Only if the Firmware Version is the latest, T3000 will update .

10

When you show the window, the tree column will show the values ,automatically.

11

show the path of a hex or bin matching with the device. clicking the blank to select the file ,the file will be checked whether the file is fit the product or not .

12

show the path of a configuration file matching with the device. clicking the blank to select the file ,the file will be checked whether the file is fit the product or not .